

The Loyola Index Family: Bounds, Parameter Geometry and Sensitivity Analysis

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Abstract

The two-parameter Gutman–Milovanović family $M_{\alpha,\beta}(G) = \sum_{uv \in E(G)} (d_u d_v)^\alpha (d_u + d_v)^\beta$ unifies the Zagreb, Randić, sum-connectivity and related degree-based indices. We study its normalized degree-imbalance exponential extension, the *Loyola index family*

$$LO(G; \alpha, \beta, \gamma) = \sum_{uv \in E(G)} (d_u d_v)^\alpha (d_u + d_v)^\beta \exp\left(\gamma \frac{|d_u - d_v|}{d_u + d_v}\right),$$

with $LO(G; \alpha, \beta, 0) = M_{\alpha,\beta}(G)$ identically.

We prove that the imbalance direction is affinely independent of the two parent directions on the degree-pair set of graphs with maximum degree at least 3, that $\log LO$ is globally convex in (α, β, γ) with gradient and Hessian given by the mean and covariance of the edge-feature vector under an induced edge measure, and we extend two-sided bounds from the parent family to $\gamma \neq 0$. On molecular graphs of maximum degree at most four the entire family is confined to the finite-dimensional space of edge-degree-pair counts;

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we quantify this ceiling exactly, exhibiting octane-isomer pairs that no vertex-degree-based index can distinguish, and showing that the pure- γ points attain the maximal degree-pair discrimination on trees of orders 7–12. A matched-budget nested leave-one-out comparison against the optimized parent family on five octane properties, repeated over one hundred random-search seeds at two budgets, finds that the free imbalance parameter improved out-of-sample prediction of the vaporization enthalpy in at least 93 of 100 seeds, usually degraded it on three properties and was mixed on the fifth (a descriptive, not inferential, finding); a pre-registered test on the 35 nonanes reproduced the vaporization result against compilation values, although ridge regression on all degree-pair counts predicts better. Hence γ is presented as a structurally interpretable and provably independent direction rather than a predictive upgrade; under the published structure-sensitivity protocol of Furtula, Gutman and Dehmer, the pure- γ points show the largest sensitivity-to-abruptness ratios in the tested comparison set.

1 Introduction

The literature on degree-based topological indices in chemical graph theory has grown substantially since the seminal Wiener index [50], the Randić index [43], and the Zagreb indices of Gutman and Trinajstić [26]. Among the many parametric extensions are the general Randić index R_α [7], the general sum-connectivity index χ_α [53], the forgotten index F revisited by Furtula and Gutman [14], the geometric–arithmetic index GA [48], the Albertson irregularity index [2] and the inverse σ -index problem [25].

While historical literature focused primarily on algebraic formulations to capture molecular branching and structural invariants, a distinct shift toward geometric interpretations occurred when Gutman introduced the Sombor index family [19, 20]. This formulation led to work such as establishing fundamental bounds [34] and evaluating how these new geometric invariants correlated with traditional degree-based metrics [49]. Much of this initial work concentrated on identifying extremal bounds within fundamental graph classes, particularly optimizing the general and standard variants of the index over molecular trees and related networks [1, 9, 11].

Contemporary studies have introduced modified metrics such as the mean Sombor index [33], the elliptic Sombor index [23], and the hyperbolic

Sombor index [6]. Concurrently, parameters like the diminished Sombor index [36,37] and the hybrid vertex-edge KG-Sombor index [30] have been developed to handle unique structural constraints. Beyond purely theoretical analysis, these descriptors are increasingly integrated into practical quantitative structure-property relationship (QSPR) models and spectral evaluations. For instance, hybrid geometric-harmonic-Zagreb descriptors are now utilized to optimize the predictive analysis of polycyclic aromatic hydrocarbons [5], while modified reverse degree invariants help characterize the graph energy and entropy of complex 2D metal-organic frameworks [28]. The diversification of structural metrics is also being studied through generalized matrix methods [54], dual coindex formulations [18], simplicial network assessments [45], and finite-dimensional linear frameworks [41]; for Sombor-type variants see the recent survey [21], and for extremal results on bond-incident-degree (BID) indices in general see [3].

The natural two-parameter unification of the product-based and sum-based degree indices is the *Gutman–Milovanović family*, introduced by Gutman, Milovanović and Milovanović [24] and studied intensively in the recent bounds and QSPR literature [17,35,39]:

$$M_{\alpha,\beta}(G) = \sum_{uv \in E(G)} (d_u d_v)^\alpha (d_u + d_v)^\beta, \quad \alpha, \beta \in \mathbb{R}.$$

Its special cases (most of them catalogued in [35]) include $M_{0,1} = M_1$ (first Zagreb), $M_{1,0} = M_2$ (second Zagreb), $M_{0,2} = HM$ (hyper-Zagreb [46]), $M_{-1,0} = {}^mM_2$ (modified second Zagreb), $M_{-1/2,0} = R$ (Randić), $M_{0,-1/2} = \chi$ (sum-connectivity [52]), $2M_{0,-1} = H$ (harmonic), $M_{1,-1} = ISI$ (inverse sum indeg), $2M_{1/2,-1} = GA$ (geometric-arithmetic), $\frac{1}{2}M_{-1/2,1} = AG$ (arithmetic-geometric), and the whole general Randić family $R_\alpha = M_{\alpha,0}$ [7] and general sum-connectivity family $\chi_\beta = M_{0,\beta}$ [53] as parameter axes. Parameter-optimized versions of this family and of related generalized descriptors have already been fitted to octane properties [4]. We claim no novelty for this backbone, for its reduction catalogue, or for its generic tunability: those belong to the Gutman–Milovanović literature.

In this paper we study a three-parameter extension of $M_{\alpha,\beta}$, the *Loy-*

ola index family, obtained by attaching a separately parameterized exponential factor in the normalized degree imbalance $q_{uv} = |d_u - d_v|/(d_u + d_v)$ of each edge: $LO(G; \alpha, \beta, \gamma) = \sum_{uv} (d_u d_v)^\alpha (d_u + d_v)^\beta e^{\gamma q_{uv}}$, so that $LO(G; \alpha, \beta, 0) = M_{\alpha, \beta}(G)$ identically. Positive γ exponentially upweights degree-imbalanced edges, negative γ downweights them, $q_{uv} = 0$ on equal-degree edges, and on regular graphs γ has no effect. The questions this paper answers are therefore about γ alone: is the imbalance direction mathematically independent of the parent parameters, what analytic structure does it carry, and does it add measurable structural or predictive information beyond an optimized Gutman–Milovanović parent?

On the mathematical side, we prove that the answer to independence is affirmative on the admissible degree-pair set of graphs of maximum degree at least three (Proposition 2; on an individual graph γ can be absorbed by the parent parameters, Remark 3), establish global log-convexity of the family in (α, β, γ) with an exact mean/covariance interpretation of its derivatives (Theorem 3), give closed-form values on standard graph classes, and derive two-sided bounds that extend inequalities known for $M_{\alpha, \beta}$ [17, 35] to the $\gamma \neq 0$ regime, several sharp in their equality cases. On the chemometric side we quantify the exact information ceiling of the entire vertex-degree-based class on molecular graphs, run a matched-budget nested comparison of LO against its optimized parent, and evaluate the family under the published structure-sensitivity protocol of Furtula, Gutman and Dehmer [15]; no universal ranking or predictive-superiority claim is made.

The distinguishing feature of the Loyola family within the degree-based BID class is the multiplicative $\exp(\gamma q_{uv})$ factor, where $q_{uv} = |d_u - d_v|/(d_u + d_v) \in [0, 1]$ is the normalized degree imbalance. Writing the summand as $\exp(\alpha \ln(d_u d_v) + \beta \ln(d_u + d_v) + \gamma q_{uv})$ shows that the family is a three-parameter subfamily of the exponential vertex-degree-based indices introduced by Rada [40] and studied further in extremal settings by Sigarreta [47]; general sufficient conditions for tree-extremal behaviour of such exponential edge-weight sums are known [16] (see also [8, 10]), bounds for general edge-weight functions over graphs of given order and size are given in [27], and no extremal-ordering novelty is claimed here. The pub-

lished one-parameter exponential indices (e.g. the exponential geometric–arithmetic [12] and exponential inverse symmetric division deg [13] indices) place their parameter on a single classical generator inside the exponent; the scoped contribution of this family is the multi-parameter *linear combination* in the exponent in which the two Gutman–Milovanović basis terms act as the factors $(d_u d_v)^\alpha (d_u + d_v)^\beta$ while the imbalance coordinate carries its own parameter γ . The same imbalance ratio appears in many published indices: the diminished Sombor index, whose edge weight is $\sqrt{(1 + q_{uv}^2)/2}$ [37]; the irregularity index $IRLA(G) = 2 \sum_{uv} q_{uv}$ [44], of which the pure- γ point $LO(G; 0, 0, \gamma) = \sum_{uv} e^{\gamma q_{uv}}$ is the exponential index in the sense of [40]; degree-ratio indices [22, 38]; and the parent family itself, since $1 - q_{uv}^2 = 4d_u d_v / (d_u + d_v)^2$ gives $M_{\alpha, -2\alpha}(G) = 4^{-\alpha} \sum_{uv} (1 - q_{uv}^2)^\alpha$ (the variable geometric–arithmetic direction). The parent family is therefore not blind to imbalance; what is new is a coordinate *linear* in q_{uv} in the exponent, which, unlike $\ln(1 - q_{uv}^2)$, is not an affine function of $\ln(d_u d_v)$ and $\ln(d_u + d_v)$ (Proposition 2). To our knowledge, and within the scope of the literature search documented in the reproducibility package, no published index couples it multiplicatively, with its own parameter, to the Gutman–Milovanović family. Since $0 \leq q_{uv} < 1$, the factor lies in $[1, e^\gamma)$ for $\gamma > 0$, equals 1 for $\gamma = 0$, and lies in $(e^\gamma, 1]$ for $\gamma < 0$; it reduces to 1 on regular graphs. The exponential form is a structural choice rather than one derived from a physical model: it lets γ act multiplicatively and only on irregularity, vanishing on regular graphs.

Also, it is useful to demonstrate the descriptor-space ceiling in which the entire family lives. On hydrogen-suppressed molecular graphs of maximum degree at most 4, every BID index is a linear functional of the ten unordered edge-degree-pair counts $m_{ij}(G) = |\{uv \in E(G) : \{d_u, d_v\} = \{i, j\}\}|$ for $1 \leq i \leq j \leq 4$; this finite-dimensional view of vertex-degree-based indices has recently been placed on a formal linear-algebraic footing by Rada [41]. Thus the Loyola family, like every BID family on this molecular class, lies inside a fixed ten-dimensional descriptor space, no matter how its three real parameters are chosen.

In sum, the contributions of this paper are: (i) the affine independence of the imbalance direction from the two Gutman–Milovanović directions

on the admissible degree-pair set for $\Delta \geq 3$ (Proposition 2); (ii) global log-convexity of the family with exact mean/covariance formulas for its parameter derivatives (Theorem 3, a specialization of the standard convexity of log-sum-exp functions); (iii) two-sided bounds extending parent-family inequalities to $\gamma \neq 0$, several sharp; (iv) an exact quantification of the information ceiling of the whole vertex-degree-based class on molecular graphs, including octane-isomer pairs indistinguishable by every such index (Proposition 4) and floor-attainment of the pure- γ points across tree orders 7–12; and (v) an honest matched-budget nested assessment of whether γ adds out-of-sample predictive information beyond the optimized parent (in a 100-seed analysis on one fixed 18-molecule dataset it improved prediction of ΔH_{vap} in every seed, and in 93–96 of 100 when the one isomer with an unresolved reference state is excluded, usually hurt on three of the other four properties, and was budget-dependent on the fifth; this describes optimizer variance over candidate streams, not inferential evidence); (vi) a pre-registered external test on the 35 nonane isomers, in which the vaporization-enthalpy effect replicates against compilation values (95–98 of 100 seeds) and the experimentally grounded boiling-point test is mixed, together with stronger baselines, including ridge regression on all degree-pair counts, which predicts better than every index model there (Section 5.4); together with the family’s behaviour under the published structure-sensitivity protocol [15], validated against published decane-class values [6]. The family is thus offered not as a universally superior predictor but as a controlled, analytically transparent probe of the BID descriptor space whose third direction is provably independent of its established parent.

2 Preliminaries

2.1 Notation

Let $G = (V, E)$ be a finite simple connected graph of order $n = |V|$ and size $m = |E|$. For a vertex $v \in V$, the degree $d_v = d_G(v)$ counts the edges incident to v . The maximum and minimum degrees are denoted

$\Delta = \Delta(G)$ and $\delta = \delta(G)$. In the chemometric sections we work with hydrogen-suppressed molecular graphs, where the assumption $\Delta \leq 4$ holds for alkane carbon skeletons (all graphs analysed here) and for most organic molecules, but not universally: hypervalent centres such as phosphorus in PCl_5 (degree 5) or sulfur in pentafluorosulfanyl groups (degree 6) exceed it.

The classical graph families used throughout are: the complete graph K_n , the cycle C_n , the path P_n , the star S_n , the complete bipartite graph $K_{p,q}$, the hypercube Q_k (the k -dimensional binary hypercube; 2^k vertices, $k \cdot 2^{k-1}$ edges, k -regular), the wheel W_n (an $(n-1)$ -cycle plus a universal vertex), the friendship graph F_p (p triangles sharing a common vertex), and the r -regular graphs.

For an edge $e = uv \in E(G)$ we write (d_u, d_v) for its *degree pair* understood as an unordered multiset $\{d_u, d_v\}$.

2.2 Definition

Definition 1 (Loyola index family). Let $G = (V, E)$ be a simple connected graph. For $\alpha, \beta, \gamma \in \mathbb{R}$, the *Loyola index family* is defined as,

$$LO(G; \alpha, \beta, \gamma) = \sum_{uv \in E(G)} (d_u d_v)^\alpha (d_u + d_v)^\beta \exp\left(\gamma \frac{|d_u - d_v|}{d_u + d_v}\right) \quad (1)$$

Equation (1) is well defined for all $\alpha, \beta \in \mathbb{R}$ on graphs without isolated vertices (so that $d_u, d_v \geq 1$ on every edge). The expression is invariant under permutations of (u, v) because $d_u d_v = d_v d_u$, $d_u + d_v = d_v + d_u$, and $|d_u - d_v| = |d_v - d_u|$; hence $LO(G; \alpha, \beta, \gamma)$ depends only on the multiset of unordered edge degree pairs of G . When $\gamma = 0$ the family reduces identically to the Gutman–Milovanović family [24, 35],

$$LO(G; \alpha, \beta, 0) = M_{\alpha, \beta}(G) = \sum_{uv \in E(G)} (d_u d_v)^\alpha (d_u + d_v)^\beta,$$

whose classical special cases are catalogued in the Introduction. Throughout, $q_{uv} = \text{imb}(d_u, d_v)$, where $\text{imb}(x, y) = |x - y|/(x + y)$, denotes the normalized degree imbalance of the edge uv ; positive γ exponentially up-

weights degree-imbalanced edges, negative γ downweights them, and on regular graphs every $q_{uv} = 0$ so that γ has no effect and $LO(G; \alpha, \beta, \gamma) = M_{\alpha, \beta}(G)$ for all γ .

2.3 Closed-form values on standard graphs

Proposition 1. *Let $\alpha, \beta, \gamma \in \mathbb{R}$. Then:*

1. $LO(K_n; \alpha, \beta, \gamma) = \frac{n(n-1)}{2} 2^\beta (n-1)^{2\alpha+\beta}$ for $n \geq 2$.

2. $LO(C_n; \alpha, \beta, \gamma) = n \cdot 4^{\alpha+\beta}$ for $n \geq 3$.

3. $LO(P_n; \alpha, \beta, \gamma) = \begin{cases} 2^\beta, & n = 2, \\ 2 \cdot 2^\alpha 3^\beta e^{\gamma/3}, & n = 3, \\ 2 \cdot 2^\alpha 3^\beta e^{\gamma/3} + (n-3) \cdot 4^{\alpha+\beta}, & n \geq 4. \end{cases}$

4. $LO(S_n; \alpha, \beta, \gamma) = (n-1)(n-1)^\alpha n^\beta \exp\left(\gamma \frac{n-2}{n}\right)$ for $n \geq 2$.

5. $LO(K_{p,q}; \alpha, \beta, \gamma) = pq(pq)^\alpha (p+q)^\beta \exp\left(\gamma \frac{|p-q|}{p+q}\right)$.

6. $LO(Q_k; \alpha, \beta, \gamma) = k \cdot 2^{k-1} \cdot 2^\beta k^{2\alpha+\beta}$ for $k \geq 1$.

7. For $n \geq 5$, W_n has $n-1$ spoke edges with degree pair $(3, n-1)$ and $n-1$ rim edges with degree pair $(3, 3)$, so

$$LO(W_n; \alpha, \beta, \gamma) = (n-1) \left[3^\alpha (n-1)^\alpha (n+2)^\beta \exp\left(\gamma \frac{n-4}{n+2}\right) + 9^\alpha 6^\beta \right].$$

8. For $p \geq 1$, F_p has $2p$ spoke edges $(2, 2p)$ and p outer edges $(2, 2)$, so

$$LO(F_p; \alpha, \beta, \gamma) = p \cdot 4^{\alpha+\beta} + 2p(4p)^\alpha (2p+2)^\beta \exp\left(\gamma \frac{p-1}{p+1}\right).$$

9. For an r -regular graph G with m edges, $LO(G; \alpha, \beta, \gamma) = m \cdot 2^\beta r^{2\alpha+\beta}$.

Proof. Each case is obtained by enumerating the edge degree pairs of the graph and applying Definition 1. For r -regular graphs and for K_n, C_n, Q_k (all regular), every edge has $d_u = d_v = r$, so $\text{imb} = 0$ and $(d_u d_v)^\alpha (d_u + d_v)^\beta = r^{2\alpha} (2r)^\beta = 2^\beta r^{2\alpha+\beta}$. Items (1), (2), (6), and (9) follow immediately. For S_n all $n-1$ edges have degree pair $(1, n-1)$ with $\text{imb} = (n-2)/n$;

item (4) follows. For $K_{p,q}$ all pq edges have degree pair (p, q) ; item (5) follows. For P_n ($n \geq 4$) there are exactly two pendant edges with pair $(1, 2)$ and $n - 3$ interior edges with pair $(2, 2)$; item (3) follows by direct summation. The boundary cases $n = 2, 3$ are immediate. Items (7) and (8) are obtained by partitioning the edges of W_n and F_p into their two edge-degree-pair classes respectively and summing. ■

3 Parameter geometry and the role of the imbalance parameter

This section establishes the two structural facts that justify introducing the third parameter at all: the imbalance direction cannot be reproduced by the parent parameters, and the family has a globally convex logarithm whose derivatives have an exact probabilistic meaning. Throughout, write $z_{uv} = (\ln(d_u d_v), \ln(d_u + d_v), q_{uv}) \in \mathbb{R}^3$ for the feature vector of the edge uv and $\theta = (\alpha, \beta, \gamma)$, so that $LO_G(\theta) = \sum_{uv \in E(G)} e^{\theta^\top z_{uv}}$.

Proposition 2 (Affine independence of the imbalance direction). *Let $\Delta \geq 3$ and let $P_\Delta = \{(i, j) : 1 \leq i \leq j \leq \Delta\}$ be the admissible degree pairs. Then the augmented feature vectors $(1, \ln(ij), \ln(i + j), q_{ij})$, $(i, j) \in P_\Delta$, span $\mathbb{R}^4$. Consequently, if two parameter triples θ, θ' satisfy $e^{\theta^\top z} = c e^{\theta'^\top z}$ for some constant $c > 0$ and all $(i, j) \in P_\Delta$, then $\theta = \theta'$ and $c = 1$; in particular no choice of (α', β') and scaling constant reproduces the action of $\gamma \neq 0$ on all admissible degree pairs. For $\Delta = 2$ the span has dimension 3, and for $\Delta = 1$ (the single pair $(1, 1)$, augmented vector $(1, 0, \ln 2, 0)$) it has dimension 1; in both cases the conclusion fails.*

Proof. It suffices to show that q_{ij} is not an affine function of $(\ln(ij), \ln(i + j))$ on P_Δ ; the four pairs $(1, 1), (2, 2), (1, 2), (1, 3)$, all admissible once $\Delta \geq 3$, witness this. Suppose $q_{ij} = a + b \ln(ij) + c \ln(i + j)$ on these pairs. From $(1, 1)$ and $(2, 2)$, where $q = 0$: $a + c \ln 2 = 0$ and $a + (2b + 2c) \ln 2 = 0$, giving $a = -c \ln 2$ and $b = -c/2$. Substituting into the $(1, 2)$ and $(1, 3)$ equations gives $\frac{1}{3} = c(\ln 3 - \frac{3}{2} \ln 2)$ and $\frac{1}{2} = c(\ln 2 - \frac{1}{2} \ln 3)$; in particular $c \neq 0$. Three times the first minus twice the second gives $0 = c(4 \ln 3 - \frac{13}{2} \ln 2)$, hence $8 \ln 3 = 13 \ln 2$, i.e. $3^8 = 2^{13}$, which is false ($6561 \neq 8192$).

Hence the augmented vectors have rank 4. If $e^{\theta^\top z} = c e^{\theta'^\top z}$ on P_Δ , then $(\theta - \theta')^\top z_{ij} = \ln c$ for all admissible pairs, i.e. $(\ln c, \theta' - \theta)$ is orthogonal to every augmented vector, so $\theta = \theta'$ and $c = 1$. For $\Delta = 2$ only the pairs $(1, 1), (1, 2), (2, 2)$ exist and a direct computation shows the augmented vectors have rank 3; for $\Delta = 1$ the single augmented vector has rank 1. $\blacksquare$

Remark (Independence versus per-graph identifiability). Proposition 2 is a statement about the edge-weight *function* on the full admissible pair set, not about every individual graph: on a single graph G , the parameters are identifiable from the *function* $\theta \mapsto LO(G; \theta)$ (never from a single value of it, which is one scalar) up to the scaling noted above only when the degree pairs actually realised by G carry four affinely independent augmented feature vectors. On a regular graph nothing about θ is identifiable beyond one linear functional; on a graph whose edges realise at most three affinely independent augmented vectors, some direction of θ — possibly the γ direction — is unidentifiable from $LO(G; \cdot)$ alone. The proposition guarantees that no such degeneracy is forced on the whole molecular class $\Delta \geq 3$: the family separates parameters as a family of descriptors, even though particular graphs may not.

The next result is the standard convexity of a log-sum-exp (log-partition) function, whose gradient and Hessian are the mean and covariance of the exponent’s feature vector under the induced weights; we state it for LO because these moments have a direct interpretation here and because strictness depends on the degree pairs that G realises.

Theorem 3 (Global log-convexity). *Fix a graph G with $m \geq 1$ edges and define $F_G(\theta) = \log LO_G(\theta)$ on $\mathbb{R}^3$. Let $\mathbb{E}_\theta$ and Cov_θ denote mean and covariance under the edge probability measure $w_{uv}(\theta) = e^{\theta^\top z_{uv}} / LO_G(\theta)$. Then F_G is convex, with*

$$\nabla F_G(\theta) = \mathbb{E}_\theta[z], \quad \nabla^2 F_G(\theta) = \text{Cov}_\theta(z) \succeq 0.$$

In particular $\partial_\gamma F_G = \mathbb{E}_\theta[q] \in [0, 1)$ and $\partial_\gamma^2 F_G = \text{Var}_\theta(q) \geq 0$, so $\log LO$ is nondecreasing and convex in γ , strictly increasing in γ unless every

edge of G joins vertices of equal degree. Moreover $\text{Var}_\theta(q) = 0$ at some (equivalently, every) θ if and only if q_{uv} is constant over the edges of G : this covers regular graphs ($q \equiv 0$), but also, e.g., (Δ, δ) -biregular graphs, on which every edge carries the same nonzero imbalance and $\log LO$ is affine and strictly increasing in γ . F_G is strictly convex if and only if the edge feature vectors $\{z_{uv}\}_{uv \in E}$ do not lie in a proper affine subspace of $\mathbb{R}^3$; on a regular graph $F_G(\theta) = \log m + \theta^\top z_0$ is affine and $\text{Cov}_\theta(z) = 0$.

Proof. Direct differentiation of $LO_G(\theta) = \sum_{uv} e^{\theta^\top z_{uv}}$ gives $\partial_k F_G = \sum_{uv} w_{uv} z_{uv,k} = \mathbb{E}_\theta[z_k]$ and $\partial_k \partial_l F_G = \mathbb{E}_\theta[z_k z_l] - \mathbb{E}_\theta[z_k] \mathbb{E}_\theta[z_l]$, the (k, l) -entry of $\text{Cov}_\theta(z)$, which is positive semidefinite; convexity follows. The measure $w(\theta)$ has full support on E for every θ , so $\text{Cov}_\theta(z)$ is singular in a direction v iff $v^\top z_{uv}$ is constant over all edges, i.e. iff the z_{uv} lie in a proper affine subspace; this proves the strictness criterion. Since $q_{uv} \in [0, 1]$, $\mathbb{E}_\theta[q] \in [0, 1]$, and $\mathbb{E}_\theta[q] = 0$ iff $q_{uv} = 0$ on every edge. On an r -regular graph all z_{uv} equal $z_0 = (2 \ln r, \ln 2r, 0)$. ■

Theorem 3 places the family in the classical exponential-family framework: moving γ reweights the edge measure toward imbalanced edges at rate $\mathbb{E}_\theta[q]$, and the Cauchy–Schwarz midpoint inequality of Section 4 is precisely midpoint log-convexity, i.e. a corollary of this theorem.

Proposition 4 (Degree-pair information ceiling). *If two graphs G and H have identical unordered edge-degree-pair count vectors, $m_{ij}(G) = m_{ij}(H)$ for all $i \leq j$, then $LO(G; \theta) = LO(H; \theta)$ for every $\theta \in \mathbb{R}^3$, and more generally $TI(G) = TI(H)$ for every vertex-degree-based index TI . Among the 18 octane isomers exactly 16 distinct count vectors occur: the pairs $\{3\text{-methylheptane}, 4\text{-methylheptane}\}$ and $\{3,4\text{-dimethylhexane}, 3\text{-ethyl-2-methylpentane}\}$ share their count vectors while differing in measured properties (Table 2), so every vertex-degree-based descriptor, tuned or not, carries a nonzero irreducible error on this benchmark.*

Proof. By Definition 1, $LO(G; \theta)$ depends on G only through the multiset of unordered edge degree pairs, i.e. through $(m_{ij}(G))_{i \leq j}$; the same holds for every vertex-degree-based index [41]. The counts for the 18 octane isomers are computed and cross-checked in the reproducibility package; both collision pairs have their common count vectors listed in Table 2. ■

The ceiling is tight in the following sense: distinct count vectors are separated by the family at almost every parameter choice.

Proposition 5 (Generic discrimination of distinct profiles). *Fix $\Delta \geq 1$ and let $m \neq m'$ be two distinct edge-degree-pair count vectors supported on the admissible pairs P_Δ . Then $D(\theta) = \sum_{(i,j) \in P_\Delta} (m_{ij} - m'_{ij}) e^{\theta^\top z_{ij}}$ is a real-analytic function of θ that is not identically zero, and its zero set has Lebesgue measure zero in $\mathbb{R}^3$. Consequently, any two graphs with distinct count vectors are distinguished by $LO(\cdot; \theta)$ for almost every $\theta \in \mathbb{R}^3$; only graphs with identical count vectors (Proposition 4) are indistinguishable for all θ .*

Proof. The map $(i, j) \mapsto z_{ij}$ is injective on P_Δ , since $\ln(ij)$ and $\ln(i+j)$ recover the product and the sum, which determine the unordered pair. Choose $v \in \mathbb{R}^3$ such that the inner products $v^\top z_{ij}$ are pairwise distinct (this excludes finitely many hyperplanes of v 's). Along the ray $\theta = tv$, $D(tv) = \sum_k c_k e^{t\lambda_k}$ with distinct exponents λ_k and coefficients $c_k = m_{i_k j_k} - m'_{i_k j_k}$, not all zero; such an exponential sum vanishes identically only if all $c_k = 0$, so $D \not\equiv 0$ on the ray and hence on $\mathbb{R}^3$. D is a finite sum of exponentials of linear functionals, hence real-analytic on $\mathbb{R}^3$, and the zero set of a nonzero real-analytic function has Lebesgue measure zero. ■

The proof uses only the injectivity of $(i, j) \mapsto (\ln(ij), \ln(i+j))$, so the same conclusion already holds for the parent family $M_{\alpha, \beta}$ on the (α, β) -plane. Proposition 5 is therefore a property the Loyola family inherits, not an advantage conferred by γ . Likewise, since $q_{ij} = \sqrt{1 - 4ij/(i+j)^2}$ is a function of the pair, the imbalance coordinate adds no graph information beyond the edge-degree-pair counts (every vertex-degree-based index is a functional of those counts [41]); what γ adds is the new functional direction of Proposition 2, i.e. a different weighting of the same information.

4 Bounds for the Loyola family

This section establishes a coherent set of upper and lower bounds on $LO(G; \alpha, \beta, \gamma)$. The development proceeds from a bracket bound via the

edge-contribution function ψ , bounds in (n, m, Δ, δ) for the non-negative parameter region, Cauchy–Schwarz and Jensen-type bounds in the first two Zagreb indices M_1 and M_2 , a uniform ratio-bound theorem, and bounds via the generalised Randić, sum-connectivity, and Sombor indices. Throughout, G is a simple connected graph with $\delta \leq d_v \leq \Delta$ for every $v \in V(G)$. The inequalities of Propositions 8 and 9 and of Theorems 7 and 15, the convexity part of Theorem 3, the rank-4 statement of Proposition 2 for $\Delta \geq 3$, and the order-13 profile collision of Section 5.5 have additionally been machine-checked in Lean 4 with Mathlib (graphs modelled by their multisets of edge degree pairs; equality characterizations not formalized); the formalization accompanies the reproducibility package.

Remark (Attribution at $\gamma = 0$). At $\gamma = 0$ the family coincides with $M_{\alpha, \beta}$, for which inequalities of all the types developed here are already known: (Δ, δ, m) -bounds with regular/biregular equality and two-sided comparisons against M_1 , M_2 , R , χ , H , GA , AG and ISI appear in [35], and Jensen- and Hölder-type relations in [17] and further inequalities in [39]. The $\gamma = 0$ specializations of the results below are therefore not claimed as new; the content of this section is their extension to $\gamma \neq 0$, where the imbalance factor $e^{\gamma q_{uv}}$ enters every edge weight, and the tracking of how γ deforms the equality cases. Theorems 10–12 and Propositions 13–14 are stated at $\gamma = 0$ only; they are standard Jensen and Cauchy–Schwarz consequences for $M_{\alpha, \beta}$, included to make the comparison set self-contained, and are not claimed as new.

The summand in (1) depends only on the unordered edge degree pair (d_u, d_v) . We isolate this dependence as follows:

Definition 2. For real parameters α, β, γ and positive reals x, y ,

$$\psi(x, y; \alpha, \beta, \gamma) = (xy)^\alpha (x + y)^\beta \exp\left(\gamma \frac{|x - y|}{x + y}\right). \quad (2)$$

With this notation $LO(G; \alpha, \beta, \gamma) = \sum_{uv \in E(G)} \psi(d_u, d_v; \alpha, \beta, \gamma)$.

Proposition 6. *Let G be a simple connected graph of size $m \geq 1$ with degrees in $[\delta, \Delta]$. Then*

$$m \cdot \psi_{\min}(\delta, \Delta; \alpha, \beta, \gamma) \leq LO(G; \alpha, \beta, \gamma) \leq m \cdot \psi_{\max}(\delta, \Delta; \alpha, \beta, \gamma), \quad (3)$$

where, $\psi_{\min} = \min\{\psi(i, j; \alpha, \beta, \gamma) : \delta \leq i \leq j \leq \Delta\}$ and $\psi_{\max} = \max\{\psi(i, j; \alpha, \beta, \gamma) : \delta \leq i \leq j \leq \Delta\}$. The upper inequality is an equality if and only if every edge of G has a degree pair attaining $\psi_{\max}$, and the lower inequality is an equality if and only if every edge has a degree pair attaining $\psi_{\min}$.

Proof. Because G has $\delta \leq d_v \leq \Delta$ for every $v \in V(G)$, every edge $uv \in E(G)$ has its degree pair $\{d_u, d_v\}$ in the admissible set $\mathcal{P}(\delta, \Delta) := \{(i, j) \in \mathbb{Z}^2 : \delta \leq i \leq j \leq \Delta\}$. By the definitions of $\psi_{\min}$ and $\psi_{\max}$ as the minimum and maximum of ψ over $\mathcal{P}(\delta, \Delta)$,

$$\psi_{\min}(\delta, \Delta; \alpha, \beta, \gamma) \leq \psi(d_u, d_v; \alpha, \beta, \gamma) \leq \psi_{\max}(\delta, \Delta; \alpha, \beta, \gamma).$$

Summing each of the two per-edge inequalities over the m edges of G and using $\sum_{uv \in E(G)} \psi(d_u, d_v; \alpha, \beta, \gamma) = LO(G; \alpha, \beta, \gamma)$ gives

$$m \cdot \psi_{\min}(\delta, \Delta; \alpha, \beta, \gamma) \leq LO(G; \alpha, \beta, \gamma) \leq m \cdot \psi_{\max}(\delta, \Delta; \alpha, \beta, \gamma),$$

For the equality cases, the upper inequality $LO(G; \alpha, \beta, \gamma) \leq m \cdot \psi_{\max}$ is the sum of m per-edge inequalities $\psi(d_u, d_v; \alpha, \beta, \gamma) \leq \psi_{\max}$; this sum is an equality if and only if every per-edge inequality is, equivalently every edge uv has a degree pair attaining $\psi_{\max}$ over $\mathcal{P}(\delta, \Delta)$. The lower equality case is the dual. ■

Theorem 7. *Let $\alpha_1 + \alpha_2 = 2\alpha$, $\beta_1 + \beta_2 = 2\beta$, and $\gamma_1 + \gamma_2 = 2\gamma$. Then for every simple connected graph G without isolated vertices,*

$$LO(G; \alpha, \beta, \gamma) \leq \sqrt{LO(G; \alpha_1, \beta_1, \gamma_1) \cdot LO(G; \alpha_2, \beta_2, \gamma_2)}$$

Equality holds if and only if the ratio

$$\psi(d_u, d_v; \alpha_1, \beta_1, \gamma_1) / \psi(d_u, d_v; \alpha_2, \beta_2, \gamma_2)$$

is the same for every edge of G .

Proof. For each edge $e = uv \in E(G)$ and each $k \in \{1, 2\}$ write

$$\psi_e^{(k)} := \psi(d_u, d_v; \alpha_k, \beta_k, \gamma_k) = (d_u d_v)^{\alpha_k} (d_u + d_v)^{\beta_k} \exp\left(\gamma_k \frac{|d_u - d_v|}{d_u + d_v}\right).$$

Both $\psi_e^{(1)}$ and $\psi_e^{(2)}$ are strictly positive on every edge of a graph without isolated vertices.

By the additivity of the exponents and the hypothesis $\alpha_1 + \alpha_2 = 2\alpha$, $\beta_1 + \beta_2 = 2\beta$, $\gamma_1 + \gamma_2 = 2\gamma$,

$$\begin{aligned} \psi_e^{(1)} \psi_e^{(2)} &= (d_u d_v)^{\alpha_1 + \alpha_2} (d_u + d_v)^{\beta_1 + \beta_2} \exp\left((\gamma_1 + \gamma_2) \frac{|d_u - d_v|}{d_u + d_v}\right) \\ &= (d_u d_v)^{2\alpha} (d_u + d_v)^{2\beta} \exp\left(2\gamma \frac{|d_u - d_v|}{d_u + d_v}\right) \\ &= \psi(d_u, d_v; \alpha, \beta, \gamma)^2. \end{aligned}$$

Taking square roots gives $\sqrt{\psi_e^{(1)} \psi_e^{(2)}} = \psi(d_u, d_v; \alpha, \beta, \gamma)$ for every edge. Apply the Cauchy–Schwarz inequality to the positive sequences $a_e = \sqrt{\psi_e^{(1)}}$, $b_e = \sqrt{\psi_e^{(2)}}$ indexed over $E(G)$:

$$\sum_{e \in E(G)} a_e b_e \leq \sqrt{\sum_{e \in E(G)} a_e^2} \sqrt{\sum_{e \in E(G)} b_e^2}.$$

Substituting $a_e b_e = \sqrt{\psi_e^{(1)} \psi_e^{(2)}} = \psi(d_u, d_v; \alpha, \beta, \gamma)$, $\sum_e a_e^2 = \sum_e \psi_e^{(1)} = LO(G; \alpha_1, \beta_1, \gamma_1)$, and $\sum_e b_e^2 = \sum_e \psi_e^{(2)} = LO(G; \alpha_2, \beta_2, \gamma_2)$ gives

$$\begin{aligned} LO(G; \alpha, \beta, \gamma) &= \sum_{e \in E(G)} \psi(d_u, d_v; \alpha, \beta, \gamma) \\ &\leq \sqrt{LO(G; \alpha_1, \beta_1, \gamma_1) LO(G; \alpha_2, \beta_2, \gamma_2)}, \end{aligned}$$

which is the inequality.

The Cauchy–Schwarz inequality is an equality if and only if the sequences (a_e) and (b_e) are proportional, i.e. there exists a constant $\lambda \geq 0$ such that $b_e = \lambda a_e$ for every $e \in E(G)$, equivalently $\psi_e^{(2)}/\psi_e^{(1)} = \lambda^2$ is the same constant on every edge. ■

Proposition 8. *Let G be a connected graph of size m and maximum degree*

Δ . For $\alpha, \beta \geq 0$ and $\gamma \geq 0$,

$$LO(G; \alpha, \beta, \gamma) \leq m \cdot \Delta^{2\alpha} \cdot (2\Delta)^\beta \cdot \exp\left(\gamma \frac{\Delta - 1}{\Delta + 1}\right). \quad (4)$$

Proof. Fix an edge $uv \in E(G)$. By the definition of Δ we have $1 \leq d_u, d_v \leq \Delta$. We bound the three multiplicative factors of $\psi(d_u, d_v; \alpha, \beta, \gamma)$ separately.

First factor. Because $d_u, d_v \leq \Delta$ we obtain $d_u d_v \leq \Delta^2$. Since $\alpha \geq 0$ and the map $t \mapsto t^\alpha$ is non-decreasing on $\mathbb{R}_{>0}$, $(d_u d_v)^\alpha \leq \Delta^{2\alpha}$.

Second factor. Similarly $d_u + d_v \leq 2\Delta$, and the map $t \mapsto t^\beta$ is non-decreasing for $\beta \geq 0$, so $(d_u + d_v)^\beta \leq (2\Delta)^\beta$.

Third factor. The normalised imbalance $\text{imb}(x, y) = |x - y|/(x + y)$ is maximised over the admissible set $\{(i, j) : 1 \leq i \leq j \leq \Delta\}$ at the pair $(1, \Delta)$, where $\text{imb}(1, \Delta) = (\Delta - 1)/(\Delta + 1)$. Hence $|d_u - d_v|/(d_u + d_v) \leq (\Delta - 1)/(\Delta + 1)$ for every edge. Because $\gamma \geq 0$ and $\exp$ is non-decreasing, $\exp(\gamma |d_u - d_v|/(d_u + d_v)) \leq \exp(\gamma(\Delta - 1)/(\Delta + 1))$. Multiplying the three factor-wise bounds and summing over the m edges of G yields (4). For $\gamma > 0$ and $\Delta \geq 2$ the bound is attained if and only if $\alpha = \beta = 0$ and every edge has degree pair $(1, \Delta)$ (for connected G , $G \cong K_{1, \Delta}$); if $\alpha > 0$ or $\beta > 0$ the three factor-wise maxima are not simultaneously attainable and the inequality is strict. $\blacksquare$

Proposition 9. *Let G be a connected graph of size m and minimum degree δ . For $\alpha, \beta \geq 0$ and $\gamma \geq 0$,*

$$LO(G; \alpha, \beta, \gamma) \geq m \cdot \delta^{2\alpha} \cdot (2\delta)^\beta, \quad (5)$$

with equality if and only if G is δ -regular, except in the degenerate case $(\alpha, \beta, \gamma) = (0, 0, 0)$, where $LO(G; \alpha, \beta, \gamma) \equiv m$ and the bound holds with equality for every graph.

Proof. Fix an arbitrary edge $uv \in E(G)$. By the definition of $\delta = \delta(G)$ we have $d_u, d_v \geq \delta$, and we bound the three multiplicative factors of $\psi(d_u, d_v; \alpha, \beta, \gamma)$ separately. Because $d_u, d_v \geq \delta$ and $\alpha \geq 0$, $(d_u d_v)^\alpha \geq \delta^{2\alpha}$; similarly $(d_u + d_v)^\beta \geq (2\delta)^\beta$; and since $\gamma \geq 0$ and the imbalance is non-negative, $\exp(\gamma |d_u - d_v|/(d_u + d_v)) \geq 1$. Multiplying the three lower

bounds gives $\psi(d_u, d_v; \alpha, \beta, \gamma) \geq \delta^{2\alpha}(2\delta)^\beta$ for every edge, and summing over the m edges yields (5).

For the equality case, if $(\alpha, \beta, \gamma) = (0, 0, 0)$ then $LO(G; 0, 0, 0) = m$ and the lower bound is also m , so equality holds for every graph. Assume now that $(\alpha, \beta, \gamma) \neq (0, 0, 0)$. Equality in the summed inequality requires equality in every active per-edge factor. If $\alpha > 0$ or $\beta > 0$, this forces $d_u = d_v = \delta$ on every edge. If $\alpha = \beta = 0$ and $\gamma > 0$, equality in the exponential factor requires $d_u = d_v$ on every edge; since G is connected, this implies that G is regular, hence δ -regular. Conversely, if G is δ -regular then every edge has degree pair (δ, δ) and equality holds. Thus, outside the degenerate case $(0, 0, 0)$, equality holds if and only if G is δ -regular. ■

Theorem 10. *Let $\alpha = \gamma = 0$ and $\beta \geq 1$. Then*

$$LO(G; 0, \beta, 0) = \sum_e (d_u + d_v)^\beta \geq m^{1-\beta} M_1(G)^\beta,$$

where $M_1(G) = \sum_{v \in V} d_v^2 = \sum_{uv \in E} (d_u + d_v)$. At $\beta = 1$ both sides equal $M_1(G)$ identically; for $\beta > 1$ equality holds if and only if $d_u + d_v$ is constant on every edge of G .

Proof. Setting $\alpha = \gamma = 0$ in Definition 1 collapses the edge-contribution function to $\psi(d_u, d_v; 0, \beta, 0) = (d_u + d_v)^\beta$, so $LO(G; 0, \beta, 0) = \sum_{uv \in E(G)} (d_u + d_v)^\beta$. The function $\varphi(t) = t^\beta$ satisfies $\varphi''(t) = \beta(\beta - 1)t^{\beta-2} \geq 0$ for $\beta \geq 1$; hence φ is convex on $\mathbb{R}_{>0}$, with strict convexity for $\beta > 1$. Apply Jensen's inequality (convex case) to the $m = |E(G)|$ positive values $\{d_u + d_v : uv \in E(G)\}$ with uniform weights $1/m$:

$$\left(\frac{M_1(G)}{m} \right)^\beta = \varphi \left(\frac{1}{m} \sum_{uv} (d_u + d_v) \right) \leq \frac{1}{m} \sum_{uv} (d_u + d_v)^\beta = \frac{LO(G; 0, \beta, 0)}{m},$$

using $\sum_{uv \in E(G)} (d_u + d_v) = M_1(G)$. Multiplying both sides by $m > 0$ yields the bound. At $\beta = 1$, φ is affine and Jensen is the identity $LO(G; 0, 1, 0) = M_1(G)$; for $\beta > 1$ strict convexity gives equality if and only if $d_u + d_v$ is constant over every edge. ■

Theorem 11. *Let $\alpha = \gamma = 0$ and $0 < \beta < 1$. Then $LO(G; 0, \beta, 0) = \sum_e (d_u + d_v)^\beta \leq m^{1-\beta} M_1(G)^\beta$, with equality if and only if $d_u + d_v$ is constant on every edge of G .*

Proof. The function $t \mapsto t^\beta$ is strictly concave on $\mathbb{R}_{>0}$ for $\beta \in (0, 1)$. Jensen's inequality in the concave case applied to the values $\{d_u + d_v : uv \in E(G)\}$ gives $\frac{1}{m} \sum_e (d_u + d_v)^\beta \leq (M_1(G)/m)^\beta$, using $M_1(G) = \sum_{uv} (d_u + d_v)$. Multiplying by m gives the upper bound; strict concavity gives equality if and only if $d_u + d_v$ is constant on every edge. ■

Theorem 12. *Let $\beta = \gamma = 0$ and $\alpha \geq 1$. Then*

$$LO(G; \alpha, 0, 0) = \sum_{uv \in E(G)} (d_u d_v)^\alpha = R_\alpha(G) \geq m^{1-\alpha} M_2(G)^\alpha,$$

where $M_2(G) = \sum_{uv \in E(G)} d_u d_v$ is the second Zagreb index and $R_\alpha(G) = \sum_{uv \in E(G)} (d_u d_v)^\alpha$ is the general Randić index. At $\alpha = 1$ both sides equal $M_2(G)$ identically; for $\alpha > 1$ equality holds if and only if the product $d_u d_v$ is constant on every edge of G .

Proof. Setting $\beta = \gamma = 0$ collapses the edge contribution to $(d_u d_v)^\alpha$, so $LO(G; \alpha, 0, 0) = R_\alpha(G)$. The function $\varphi(t) = t^\alpha$ has $\varphi''(t) = \alpha(\alpha - 1)t^{\alpha-2} \geq 0$ for $\alpha \geq 1$ and is convex, strictly for $\alpha > 1$. Jensen (convex case) on $\{d_u d_v : uv \in E(G)\}$ with weights $1/m$ gives $(M_2(G)/m)^\alpha \leq R_\alpha(G)/m$, using $\sum_{uv} d_u d_v = M_2(G)$. Multiplying by m yields the bound; $\alpha = 1$ is the affine identity, and for $\alpha > 1$ strict convexity gives equality if and only if $d_u d_v$ is constant on every edge. ■

Proposition 13. *Let $\gamma = 0$, and define $R_a(G) = \sum_{uv \in E(G)} (d_u d_v)^a$ and $\chi_b(G) = \sum_{uv \in E(G)} (d_u + d_v)^b$ (so $R = R_{-1/2}$ and $\chi = \chi_{-1/2}$). Then for every graph G ,*

$$LO(G; \alpha, \beta, 0) \leq \sqrt{R_{2\alpha}(G) \cdot \chi_{2\beta}(G)}. \quad (6)$$

In particular, at $(\alpha, \beta) = (-1/4, -1/4)$ this specialises to

$$LO(G; -1/4, -1/4, 0) \leq \sqrt{R(G) \chi(G)}.$$

Proof. Apply Theorem 7 with $(\alpha_1, \beta_1, \gamma_1) = (2\alpha, 0, 0)$ and $(\alpha_2, \beta_2, \gamma_2) = (0, 2\beta, 0)$, whose components sum to $(2\alpha, 2\beta, 0)$, meeting the hypotheses since $\gamma = 0$. By Definition 1, $LO(G; 2\alpha, 0, 0) = R_{2\alpha}(G)$ and $LO(G; 0, 2\beta, 0) = \chi_{2\beta}(G)$, so Theorem 7 gives (6). The equality case is that of Theorem 7: the ratio $(d_u d_v)^{2\alpha} / (d_u + d_v)^{2\beta}$ is constant over every edge. ■

Proposition 14. *For $\alpha = 0$, $0 < \beta < 1$, and $\gamma = 0$, the Loyola quantity $LO(G; 0, \beta, 0) = \chi_\beta(G)$ satisfies*

$$LO(G; 0, \beta, 0) \leq m^{1-\beta} \cdot (\sqrt{2} SO(G))^\beta, \quad (7)$$

where $SO(G) = \sum_{uv \in E(G)} \sqrt{d_u^2 + d_v^2}$ is the Sombor index.

Proof. For positive reals x, y , $x^2 + y^2 \geq (x + y)^2 / 2$ follows from $(x + y)^2 \leq 2(x^2 + y^2)$ by the AM–GM inequality $xy \leq (x^2 + y^2) / 2$. Taking square roots with $x = d_u, y = d_v$ and summing over $E(G)$ gives $SO(G) \geq M_1(G) / \sqrt{2}$, i.e. $M_1(G) \leq \sqrt{2} SO(G)$. For $0 < \beta < 1$, the concave Jensen bound (Theorem 11) gives $LO(G; 0, \beta, 0) \leq m^{1-\beta} M_1(G)^\beta$, and since $t \mapsto t^\beta$ is non-decreasing for $\beta > 0$, substituting $M_1(G) \leq \sqrt{2} SO(G)$ yields (7). ■

The next theorem gives a single uniform template for bounding $LO(G; \alpha, \beta, \gamma)$ both above and below in terms of any classical bond-incident-degree index whose edge contribution is strictly positive on the admissible degree pairs of the graph.

Theorem 15. *Let G be a simple graph with maximum degree at most Δ , and let $I_h(G) = \sum_{uv \in E(G)} h(d_u, d_v)$ be a BID index with edge contribution $h(i, j)$ strictly positive on every degree pair (i, j) that appears in G . Define*

$$c_{\min}^h = \min_{(i,j)} \frac{\psi(i, j; \alpha, \beta, \gamma)}{h(i, j)}, \quad c_{\max}^h = \max_{(i,j)} \frac{\psi(i, j; \alpha, \beta, \gamma)}{h(i, j)},$$

where the extrema range over the degree pairs actually appearing in G . Then

$$c_{\min}^h \cdot I_h(G) \leq LO(G; \alpha, \beta, \gamma) \leq c_{\max}^h \cdot I_h(G), \quad (8)$$

with equality on either side if and only if every edge of G attains the corresponding extremum of ψ/h .

Proof. Fix an edge $uv \in E(G)$ and define

$$r_{uv} := \psi(d_u, d_v; \alpha, \beta, \gamma) / h(d_u, d_v) > 0.$$

By definition $c_{\min}^h \leq r_{uv} \leq c_{\max}^h$, so multiplying by $h(d_u, d_v) > 0$ gives the per-edge bounds $c_{\min}^h h(d_u, d_v) \leq \psi(d_u, d_v; \alpha, \beta, \gamma)$ and $\psi(d_u, d_v; \alpha, \beta, \gamma) \leq c_{\max}^h h(d_u, d_v)$. Summing over $E(G)$ and using $\sum_{uv} h(d_u, d_v) = I_h(G)$ and $\sum_{uv} \psi = LO(G; \alpha, \beta, \gamma)$ gives (8); the equality conditions are that every edge attains the corresponding extremum of ψ/h over the appearing pairs. ■

Theorem 15 is a *sharp graph-specific* comparison: its constants depend on the degree pairs that actually appear in G . A uniform version follows immediately by enlarging the range of the extrema.

Corollary (Uniform Δ -bounded version). *With the hypotheses of Theorem 15, but with h strictly positive on all of $P_\Delta = \{(i, j) : 1 \leq i \leq j \leq \Delta\}$ and the extrema taken over P_Δ , inequality (8) holds simultaneously for every graph of maximum degree at most Δ , with constants independent of the individual graph. Equality on either side holds if and only if every edge of G realises a degree pair attaining the corresponding extremum over P_Δ ; the uniform constants are in general strictly worse than the graph-specific ones, since the extrema are taken over a superset.*

Proposition 16. *At $\gamma = 0$, $LO(G; \alpha, \beta, 0)$ coincides with the two-parameter (α, β) degree-based index $\sum_{uv \in E(G)} (d_u d_v)^\alpha (d_u + d_v)^\beta$. Let $E_{\max}(G)$ denote the set of edges realising $\text{imb}_{\max}(G) = \max_{uv} \text{imb}(d_u, d_v)$ and $E_{\min}(G)$ the set realising $\text{imb}_{\min}(G) = \min_{uv} \text{imb}(d_u, d_v)$. Then as $\gamma \rightarrow +\infty$,*

$$\frac{LO(G; \alpha, \beta, \gamma)}{e^{\gamma \text{imb}_{\max}(G)}} \longrightarrow \sum_{uv \in E_{\max}(G)} (d_u d_v)^\alpha (d_u + d_v)^\beta,$$

and as $\gamma \rightarrow -\infty$,

$$LO(G; \alpha, \beta, \gamma) / e^{\gamma \text{imb}_{\min}(G)} \longrightarrow \sum_{uv \in E_{\min}(G)} (d_u d_v)^\alpha (d_u + d_v)^\beta.$$

Proof. At $\gamma = 0$ the exponential factor is 1 on every edge. For the asymptotics, write $\mu(G) := \text{imb}_{\max}(G)$ and $\varepsilon_{uv} := \mu(G) - \text{imb}(d_u, d_v) \geq 0$, so that $LO(G; \alpha, \beta, \gamma) / e^{\gamma \mu(G)} = \sum_{uv} (d_u d_v)^\alpha (d_u + d_v)^\beta e^{-\gamma \varepsilon_{uv}}$. For $uv \in E_{\max}(G)$, $\varepsilon_{uv} = 0$ and the summand is γ -independent; for $uv \notin E_{\max}(G)$, $\varepsilon_{uv} > 0$ and $e^{-\gamma \varepsilon_{uv}} \rightarrow 0$ as $\gamma \rightarrow +\infty$. Since $E(G)$ is finite the limit is termwise, giving the identity. The $\gamma \rightarrow -\infty$ case is dual. ■

5 Sensitivity analysis

We benchmark the Loyola family against the classical indices it reduces to, on three standard testbeds, following the chemometric benchmark of the diminished Sombor study [37]: the 18 octane isomers (five NIST physico-chemical properties), the 106 non-isomorphic trees of order 10, and the 75 decane isomers. The comparison set is restricted to the exact reductions of Section 1 — $M_1, M_2, HM, {}^mM_2, R, \chi, H/2, ISI, GA, AG$ — together with the two fixed pure- γ benchmark points $LO(0, 0, 1)$ and $LO(0, 0, 2)$ chosen for this study. No parameter in this comparison is tuned against the property data — the effect of data-driven tuning is examined separately, as a negative control, in Section 5.2; the edge count $m = LO(0, 0, 0)$ is omitted as it is constant on the octanes. All comparisons in this section are descriptive; given the number of properties, indices, and tuning candidates examined, no confirmatory inference is claimed. All quantities are computed in double precision and rounded *once* for display (three decimals for correlations, Q^2 and RMSE, four for structure sensitivity); tabulated gains and differences are differences of full-precision values rounded once, so they may differ by one unit in the last place from the difference of two displayed entries.

5.1 Property correlations on the octane isomers

Table 1 reports the Pearson correlation r between each index and each of the five physicochemical properties of the 18 octane isomers (boiling point T_B , enthalpy of formation ΔH_f , enthalpy of vaporization ΔH_{vap} , entropy S , acentric factor ω ; property data traced to the NIST WebBook [32] (dataset compiled January 2025; individual values re-checked August–October 2026), except the acentric factors, which NIST does not list and which were taken from KDB-derived compilations; the same five properties form the octane benchmark used in [37]). Units are: T_B in $^{\circ}\text{C}$; ΔH_f and ΔH_{vap} in kcal mol^{-1} (the latter at 298 K); S in $\text{cal mol}^{-1}\text{K}^{-1}$; and ω dimensionless. Values for ΔH_{vap} and entropy S vary slightly across compilations; all comparisons here are made within this single, internally consistent dataset. Calculations use this author-compiled property table attributed to the NIST WebBook. All 90 molecule-level values were traced against authoritative sources (NIST WebBook first; KDB-derived compilations for the acentric factor, which NIST does not list), and classified under a strict taxonomy in which “verified” requires agreement at the value’s reported precision: 3 match exactly, 29 match a specific recorded determination at reported precision after unit conversion (gas phase confirmed for formation enthalpies and entropies), 37 agree within the audit tolerances, the source’s stated uncertainty or the span of several determinations without matching any one of them at reported precision, 13 lie within explained variation among authoritative determinations, 7 could not be verified against a reachable authoritative value (six gas-phase entropies absent from the NIST WebBook, and the 298 K vaporization enthalpy of 2,2,3,3-tetramethylbutane: this compound is a solid at 298 K (triple point 374 K); the NIST WebBook lists standard vaporization enthalpies of 42.94 and 42.91 kJ mol^{-1} (about 10.26 kcal mol^{-1}) together with sublimation enthalpies, whereas the benchmark value used here, 8.41 kcal mol^{-1} (35.19 kJ mol^{-1}), is that of the standard octane benchmark compilation and matches no listed determination, so its physical reference state is unresolved; Section 5.3 reports the ΔH_{vap} analysis with this isomer excluded), and one value is a marginal conflict: the acentric factor of 2,2,3,3-tetramethylbutane (0.247 vs. compiled 0.251); the

acentric factor of 2,2,4-trimethylpentane (0.305 vs. compiled 0.303) agrees only within tolerance. Substituting the alternative compiled acentric factors changes several of the ω correlations in Table 1 at the third decimal (e.g. M_2 moves from -0.988 to -0.986) but does not change the property-wise winning descriptor or any substantive interpretation; the full sensitivity computation and the machine-readable provenance trace accompany the reproducibility package. Four ΔH_{vap} values used in earlier versions of this study (3,3-dimethylhexane, 3-ethyl-3-methylpentane, 2,2,3-trimethylpentane and 2,3,3-trimethylpentane) disagreed with the NIST WebBook by $0.05\text{--}0.13\text{ kcal mol}^{-1}$; they have been corrected (Appendix A) — two to exact conversions of NIST determinations (8.97, 9.08) and two to values that lie within the span or stated uncertainty of the NIST determinations and equal the API Project 44 compilation at the displayed precision (8.83, 8.90) — and every result reported here uses the corrected data. Four entropies (*n*-octane, 2,2-dimethylhexane, 2,2,4- and 2,3,3-trimethylpentane) could not be matched to a primary source and differ from those of a widely used octane QSPR compilation by $0.15\text{--}2.29$ units (the 2,3,3-trimethylpentane value coincides with that of 2,2,3-trimethylpentane); substituting the compilation values changes all twelve S correlations in Table 1 at the third decimal (by at most 0.034 in $|r|$) but not the winning descriptor (HM). Apart from the four corrected vaporization enthalpies, the benchmark values are preserved as supplied for comparability.

Table 1. Pearson correlation r between the Loyola reductions, the fixed pure- γ benchmark points and the five physicochemical properties of the 18 octane isomers. Bold flags the best $|r|$ per column.

Index	T_B	ΔH_f	ΔH_{vap}	S	ω
$M_1 = LO(0, 1, 0)$	-0.718	-0.762	-0.935	-0.973	-0.970
$M_2 = LO(1, 0, 0)$	-0.497	-0.542	-0.811	-0.940	-0.988
$HM = LO(0, 2, 0)$	-0.654	-0.710	-0.903	-0.975	-0.982
${}^mM_2 = LO(-1, 0, 0)$	+0.855	+0.891	+0.925	+0.863	+0.794
$R = LO(-1/2, 0, 0)$	+0.819	+0.850	+0.958	+0.933	+0.898
$\chi = LO(0, -1/2, 0)$	+0.800	+0.832	+0.961	+0.949	+0.924
$H/2 = LO(0, -1, 0)$	+0.821	+0.849	+0.961	+0.933	+0.900
$ISI = LO(1, -1, 0)$	-0.005	-0.000	-0.383	-0.601	-0.738
$GA = 2 LO(1/2, -1, 0)$	+0.822	+0.858	+0.965	+0.941	+0.906
$AG = \frac{1}{2} LO(-1/2, 1, 0)$	-0.816	-0.859	-0.957	-0.939	-0.900
$LO(0, 0, 1)$	-0.828	-0.832	-0.984	-0.931	-0.929
$LO(0, 0, 2)$	-0.829	-0.846	-0.981	-0.939	-0.925

The best correlation per property is split among the reductions and the pure- γ points: the modified second Zagreb index $LO(-1, 0, 0)$ is strongest on T_B and ΔH_f , the hyper-Zagreb index $LO(0, 2, 0)$ on S , the second Zagreb index $LO(1, 0, 0)$ on ω , and the fixed pure- γ benchmark point $LO(0, 0, 1)$ on ΔH_{vap} ($|r| = 0.984$, with $LO(0, 0, 2)$ second at 0.981). At the exact-reduction triples the Loyola correlations coincide with the corresponding classical indices to floating-point precision, as they must. For each property, a descriptive post-selection paired bootstrap interval for the observed top-two $|r|$ difference ($B = 10000$, resampling the 18 isomers jointly, `default_rng(1)`) included zero; these intervals were not multiplicity-adjusted across the five properties and do not establish equality or equivalence. Individual bootstrap half-widths range from 0.016 to 0.096 across the headline cells (percentile bootstrap, $B = 10000$, `default_rng(0)`). The value of $LO(0, 0, 1)$ on ΔH_{vap} is a fixed benchmark point, not a tuned result. Recomputing each bolded correlation with single isomers deleted shifts $|r|$ by at most 0.027, indicating that no single isomer drives the leading correlations.

Figure 1 shows the signed- r heatmap of the compared indices against the five properties.

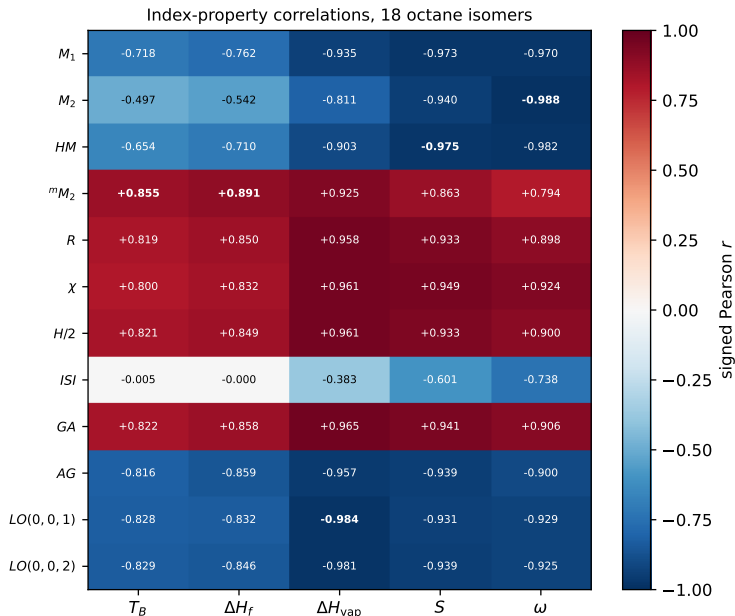


Figure 1. Signed Pearson r between the Loyola reductions, the fixed pure- γ benchmark points and the five octane physicochemical properties on the 18 isomers.

On the octanes the pure- γ points are highly collinear with the classical reductions (maximum $|r|$ with a reduction equal to 0.982 for $LO(0, 0, 1)$ and 0.993 for $LO(0, 0, 2)$).

Descriptor redundancy. The redundancy is in fact systemic, and it caps how much meaning any small $|r|$ difference in Table 1 can carry. Across the twelve compared descriptors on the 18 isomers, fourteen descriptor pairs have $|r| \geq 0.99$ (the extreme being R versus $H/2$ at $r = 0.9997$), and a principal component analysis of the standardized 18×12 descriptor matrix places 90.3% of the variance on the first component and 99.05% on the first two, with an effective rank (participation ratio of the eigenvalue spectrum) of 1.2. The twelve-index comparison set is thus essentially a two-dimensional descriptor space, and differences in the third decimal of $|r|$ between near-collinear descriptors should not be read as meaningful

rankings.

The information ceiling realised on this benchmark. Proposition 4 is not hypothetical on these data: the 18 isomers realise only 16 distinct edge-degree-pair count vectors, and the two collision pairs of Table 2 are indistinguishable by *every* vertex-degree-based index at *every* parameter choice, while their measured properties differ (by up to 1.2°C in T_B and 1.94 units in S within the first pair, and 2.1°C in T_B within the second). Every correlation in Table 1, and every tuned variant in the sections below, operates above this irreducible floor.

Table 2. The two degree-pair collision pairs among the 18 octane isomers. Each pair shares its complete edge-degree-pair count vector (shown as multiplicities of the unordered pairs), hence is indistinguishable by every vertex-degree-based index for all parameter values, yet differs in the measured properties.

Isomer	T_B	ΔH_f	ΔH_{vap}	S	ω
<i>shared count vector</i> 2(1, 2), 1(1, 3), 2(2, 2), 2(2, 3):					
3-methylheptane	118.9	-50.82	9.52	111.26	0.371
4-methylheptane	117.7	-50.69	9.48	109.32	0.372
<i>shared count vector</i> 2(1, 2), 2(1, 3), 2(2, 3), 1(3, 3):					
3,4-dimethylhexane	117.7	-50.91	9.32	106.59	0.340
3-ethyl-2-methylpentane	115.6	-50.48	9.21	106.06	0.330

5.2 Effect of parameter tuning near the best indices

The benchmark of Section 5.1 tunes no parameters: it compares only the exact reductions and the two fixed pure- γ benchmark points. It is nonetheless natural to ask whether small, data-driven adjustments of (α, β, γ) near the best-correlating indices could improve the fit. We probe this directly as a negative control. For each of the four anchors carrying the strongest column correlations in Table 1 — $M_2 = LO(1, 0, 0)$, $HM = LO(0, 2, 0)$, ${}^mM_2 = LO(-1, 0, 0)$, and the pure- γ point $LO(0, 0, 1)$ — we draw 50 random parameter triples within ± 0.3 (to two decimal places) of the anchor and correlate each against the five octane properties. The four pools are drawn, in the order listed, from one pseudo-random stream (NumPy `default_rng(42)`; each coordinate offset is a uniform draw on $[-0.3, 0.3]$)

rounded to two decimals); the percentile bootstrap uses `default_rng(0)`. Table 3 reports, for every anchor–property cell, the anchor correlation $|r|_0$, the best of the 50 perturbations in sample $|r|_{50}$, the in-sample gain Δ_{in} , the percentile-bootstrap half-width ($B = 10\,000$), the leave-one-out correlation of the anchor LOO_0 , and a *nested* leave-one-out correlation LOO_{50} in which, within each of the four globally selected anchor pools, the best of the 50 candidates is selected using the 17 training isomers of each fold only and evaluated on the held-out one, with the resulting out-of-sample gain Δ_{out} . The anchors themselves were chosen from the complete-data correlations of Table 1, and the 18 pooled fold predictions share heavily overlapping training sets, so these values are descriptive rather than independent replications.

Table 3. Decimal parameter tuning near the four best-correlating anchors on the 18 octane isomers (50 random triples within ± 0.3 of each anchor). Columns: anchor correlation $|r|_0$; best-of-50 in sample $|r|_{50}$; in-sample gain Δ_{in} ; percentile-bootstrap half-width $\pm\text{boot}$ ($B = 10000$); leave-one-out correlation of the anchor LOO_0 ; foldwise nested leave-one-out correlation of the tuned family, computed within the globally selected anchor pool, LOO_{50} ; out-of-sample gain $\Delta_{\text{out}} = \text{LOO}_{50} - \text{LOO}_0$. Gains are computed from full-precision values and rounded once, so a displayed gain may differ by one unit in the last place from the difference of the displayed operands.

Property	$ r _0$	$ r _{50}$	Δ_{in}	$\pm\text{boot}$	LOO_0	LOO_{50}	Δ_{out}
<i>near</i> $M_2 = \text{LO}(1, 0, 0)$							
T_B	0.497	0.598	+0.100	0.353	0.331	0.458	+0.127
ΔH_f	0.542	0.635	+0.093	0.358	0.395	0.540	+0.145
ΔH_{vap}	0.811	0.876	+0.066	0.185	0.768	0.838	+0.071
S	0.940	0.960	+0.021	0.067	0.919	0.943	+0.024
ω	0.988	0.995	+0.007	0.018	0.985	0.985	0.000
<i>near</i> $HM = \text{LO}(0, 2, 0)$							
T_B	0.654	0.740	+0.085	0.257	0.545	0.647	+0.102
ΔH_f	0.710	0.786	+0.076	0.255	0.645	0.741	+0.097
ΔH_{vap}	0.903	0.944	+0.041	0.082	0.871	0.921	+0.049
S	0.975	0.976	+0.002	0.027	0.967	0.960	-0.007
ω	0.982	0.989	+0.007	0.025	0.979	0.982	+0.003
<i>near</i> ${}^mM_2 = \text{LO}(-1, 0, 0)$							
T_B	0.855	0.864	+0.009	0.096	0.781	0.759	-0.021
ΔH_f	0.891	0.897	+0.006	0.087	0.865	0.837	-0.028
ΔH_{vap}	0.925	0.966	+0.041	0.068	0.900	0.953	+0.053
S	0.863	0.918	+0.055	0.188	0.812	0.881	+0.069
ω	0.794	0.881	+0.088	0.246	0.730	0.853	+0.123
<i>near</i> $\text{LO}(0, 0, 1)$							
T_B	0.828	0.835	+0.007	0.161	0.771	0.764	-0.007
ΔH_f	0.832	0.842	+0.010	0.152	0.800	0.807	+0.006
ΔH_{vap}	0.984	0.985	+0.001	0.016	0.978	0.978	-0.001
S	0.931	0.948	+0.017	0.045	0.892	0.920	+0.028
ω	0.929	0.945	+0.016	0.074	0.907	0.926	+0.019

Three patterns emerge. First, every in-sample gain Δ_{in} is smaller than the corresponding single-descriptor bootstrap uncertainty scale; this comparison is descriptive and is not a formal paired or post-selection test of the gain. Second, under the foldwise nested evaluation (within the globally selected anchor pools) these descriptive point-estimate gains do not vanish — tuning near M_2 raises the out-of-sample correlation for T_B and ΔH_f by +0.13 and +0.15, and tuning near mM_2 raises ω by +0.12 — but in every such case the anchor was a *weak* descriptor for that property, and the tuned correlation only climbs part way toward the level a strong reduction already attains (tuning near M_2 lifts T_B to 0.46, whereas mM_2

already reaches 0.78). Where the anchor is already the column winner — M_2 on ω , HM on S , $LO(0, 0, 1)$ on ΔH_{vap} — the nested gain for this pool seed is -0.0003 , -0.007 and -0.001 . These near-zero values are specific to the seed: re-drawing the pools with seeds 0–99, the nested gain for M_2 on ω is small but positive in most seeds (median about $+0.003$, positive in 96% of seeds), so the defensible statement is that tuning a column winner changes its nested correlation by at most a few thousandths. Third, tuning near the pure- γ points $LO(0, 0, 1)$ and $LO(0, 0, 2)$ does not exceed the best fixed index under nested leave-one-out on any of the five properties (Table 4). On ΔH_{vap} , where the best fixed index is the pure- γ point $LO(0, 0, 1)$ itself, the tuned value exceeds it by 0.001 in sample (0.985 vs 0.984) and falls 0.001 below it under leave-one-out (both 0.978 at three decimals; differences of full-precision values); these post-selection differences are negligible. On T_B , ΔH_f , S , and ω the best fixed reduction is higher both in and out of sample.

Table 4. Decimal tuning near the pure- γ points $LO(0, 0, 1)$ and $LO(0, 0, 2)$ versus the best fixed index in the tested comparison set for each octane property. The best fixed index is ranked by in-sample $|r|$ (its leave-one-out value in parentheses); the decimal columns give the best over two pools of 50 random triples within ± 0.3 of $LO(0, 0, 1)$ and of $LO(0, 0, 2)$, in sample and under foldwise nested leave-one-out within the globally selected pools; the $LO(0, 0, 1)$ pool is exactly the pool of Table 3, and the $LO(0, 0, 2)$ pool is the next 50 triples of the same `default_rng(42)` stream. Indices are ranked by in-sample $|r|$ rather than LOO, as the latter is inflated by degenerate fits for near-zero-correlation indices. Tuning does not exceed the best fixed index under LOO on any property; in sample it exceeds it only on ΔH_{vap} , by 0.001, a negligible post-selection difference that does not establish superiority.

Property	best fixed	$ r _{\text{fix}}$	LOO_{fix}	$ r _{\text{dec}}$	LOO_{dec}
T_B	mM_2	0.855	0.781	0.835	0.764
ΔH_f	mM_2	0.891	0.865	0.850	0.815
ΔH_{vap}	$LO(0, 0, 1)$	0.984	0.978	0.985	0.978
S	HM	0.975	0.967	0.948	0.920
ω	M_2	0.988	0.985	0.945	0.926

In short, local tuning raised several nested leave-one-out point estimates for weak anchors, and except for the 0.001 in-sample ΔH_{vap} margin

noted above did not exceed the best fixed benchmark in the tested comparison set; no formal superiority claim is made. All tabled nested values correspond to positive signed prediction–observation correlations with positive predictive Q^2 , and a constant-descriptor control confirms that absolute pooled correlation alone can be inflated for degenerate descriptors (cf. the caveat in the caption of Table 4).

5.3 Does the imbalance parameter add predictive information? A matched-budget nested comparison with the parent family

The analyses above tune around fixed anchors and are descriptive. The sharper question raised by the Gutman–Milovanović framing is direct: given identical search budgets, does freeing γ improve out-of-sample prediction beyond an optimized parent $M_{\alpha,\beta}$? We answer it with a fully nested leave-one-out design. For each property and each outer fold (one isomer held out), 200 candidate parameter vectors are drawn once from `Uniform(-2, 2)` per coordinate (triples (α, β, γ) for the Loyola model, pairs (α, β) with $\gamma = 0$ for the parent; identical budget; candidates rounded to three decimals, drawn from `NumPy default_rng(12345)` for Table 5, Loyola triples first); on the 17 training isomers only, every candidate is scored by *inner* leave-one-out root-mean-square error of the one-descriptor linear regression, the best candidate is selected, the regression is refit on all 17, and the held-out isomer is predicted. No quantity computed on the held-out isomer enters selection or fitting at any point.

The answer is mixed (Table 5): for the illustrative seed, freeing γ improves nested predictive performance on two of the five properties (T_B , $\Delta Q^2 = +0.136$, and ΔH_{vap} , $\Delta Q^2 = +0.086$) and degrades it on three (ΔH_f , S , ω), with the ΔH_f degradation driven by selection instability: the parent model selected the same (α, β) in all 18 folds there, while the three-parameter search selected widely varying triples (fold-to-fold standard deviation above 1.2 in both α and β). The selected γ values are always interior to the search box (maximum $|\gamma| = 0.48$, never within 1.5 of the boundary), so the mixed outcome is not a range artefact. On

Table 5. Matched-budget nested leave-one-out comparison of the optimized parent $M_{\alpha,\beta}$ (GM) and the optimized Loyola family (γ free) on the five octane properties. Panel (a) is a *single illustrative candidate stream* (seed 12345, budget 200): Q^2 is the outer predictive coefficient, RMSE the outer root-mean-square error, and “LO better” counts outer folds (of 18) in which the Loyola model had the strictly smaller absolute error. Panel (b) summarizes the 100-seed sweep (seeds 0–99): median ΔQ^2 with its [minimum, maximum], and the number of seeds in which $\Delta Q^2 > 0$, at budgets $B = 200$ and $B = 500$. All selection is repeated inside every outer fold; all values are full-precision results rounded once.

<i>(a) single candidate stream: seed 12345, budget 200</i>						
Property	Q^2_{GM}	Q^2_{LO}	ΔQ^2	RMSE _{GM}	RMSE _{LO}	LO better
T_B	+0.613	+0.748	+0.136	3.700	2.983	10/18
ΔH_f	+0.855	+0.549	−0.306	0.463	0.816	4/18
ΔH_{vap}	+0.884	+0.971	+0.086	0.127	0.064	13/18
S	+0.915	+0.877	−0.038	1.310	1.573	4/18
ω	+0.979	+0.956	−0.022	0.005	0.008	5/18

<i>(b) 100 candidate streams: seeds 0–99</i>				
Property	median ΔQ^2 [min, max]		seeds with $\Delta Q^2 > 0$	
	$B = 200$	$B = 500$	$B = 200$	$B = 500$
T_B	+0.010 [−0.524, +0.373]	+0.066 [−0.307, +0.231]	53/100	64/100
ΔH_f	−0.120 [−0.408, +0.206]	−0.095 [−0.373, +0.118]	14/100	24/100
ΔH_{vap}	+0.069 [+0.034, +0.102]	+0.068 [+0.024, +0.102]	100/100	100/100
S	−0.010 [−0.139, +0.037]	−0.011 [−0.099, +0.027]	35/100	29/100
ω	−0.010 [−0.074, +0.016]	−0.006 [−0.052, +0.012]	26/100	28/100

the one property where γ helps most stably, ΔH_{vap} , the identical triple $(-0.27, +0.27, +0.24)$ was selected in every fold. With $n = 18$ molecules these Q^2 differences carry substantial selection noise and we attach no inferential claim to them.

Because the comparison depends on a random candidate stream, the full nested design was repeated for one hundred seeds (0–99) at both budgets (200 and 500 candidates per model; 1000 configurations in all), together with a paired control in which the Loyola candidates are compared against the *same* (α, β) coordinates with γ set to zero, isolating the effect of γ from the independent candidate streams. Freeing γ improved nested predictive performance for ΔH_{vap} in every seed of this sweep (100/100 at both budgets, and 100/100 in the paired control; median $\Delta Q^2 = +0.069$ and $+0.068$, range $+0.024$ to $+0.102$) and *usually* degraded it for ΔH_f (86/100 and 76/100 seeds); T_B is close to an even split at budget 200 (53/100) but leans towards improvement at budget 500 (64/100), and S and ω usually degrade slightly. These counts describe optimizer variance over random candidate streams on one fixed set of 18 molecules; the molecules are never resampled, so the counts carry no inferential weight — in particular, 100/100 does not establish that γ improves ΔH_{vap} prediction for other molecules or data sets. The result is also sensitive to a single data cell: the ΔH_{vap} value of 2,2,3,3-tetramethylbutane, whose physical reference state is unresolved (Section 5.1). Repeating the complete nested comparison with that isomer excluded (17 molecules, same seeds, budgets and candidate streams), freeing γ improved ΔH_{vap} in 93/100 seeds at budget 200 and 96/100 at budget 500 (paired control 95/100 and 96/100; median $\Delta Q^2 = +0.046$ and $+0.051$, range -0.048 to $+0.098$), so the gain survives the exclusion but is no longer sign-invariant. The honest summary is therefore: at matched budget on this dataset, the imbalance parameter improved ΔH_{vap} in every seed of the sweep (93–96 of 100 without tetramethylbutane), tends to hurt on three properties (ΔH_f , S , ω), and is budget-dependent on T_B . Its established value lies in the structural properties of Section 3 and the discrimination behaviour below, not in systematic predictive gains.

5.4 External test on the nonane isomers and stronger baselines

Two limitations of the octane analysis are its size (18 molecules) and its restriction to index models. Both were addressed under a protocol pre-registered and time-stamped in the reproducibility package before any nonane value was collected: the identical nested design was applied to the 35 constitutional nonane isomers, with boiling points (35 values) and standard vaporization enthalpies (34 values; none listed for 3-ethyl-4-methylhexane) taken from the NIST WebBook [32] with every value traced to its archived source page. The two properties differ in evidential status. The boiling points are NIST averages of experimental determinations (several quoted only to 1 K), so the T_B test is experimentally grounded. Of the vaporization enthalpies, only 5 of 34 are calorimetric determinations; 29 come from two compilations, the 1961 tabulation of Labauf, Greenshields and Rossini [31], which covers all 35 nonanes and 75 decanes and is not a set of individual calorimetric measurements, and a 1971 vapour-pressure handbook [51]. Because such tabulations may rest partly on structural-increment estimates, agreement with them measures consistency with the compiled values rather than independent experimental validation, and a degree-pair-based predictor may be favoured if the compiled values were themselves derived from structural increments. The pre-registered decision rule called the γ effect *replicated* if the Loyola model beat the parent in at least 80 of 100 seeds at both budgets with positive median ΔQ^2 . Four baselines were added, each fitted entirely within the outer training fold: the training-fold mean; the best of the ten classical indices of Table 1; the irregularity index *IRLA*; and ridge regression on the full edge-degree-pair count vector, with its penalty chosen by inner leave-one-out.

On the external set the pre-registered verdicts are: for ΔH_{vap} (compilation values) the γ effect *replicates* (LO better than GM in 95/100 and 98/100 seeds, median $\Delta Q^2 = +0.048$ and $+0.053$), and for T_B it is *mixed* (58/100 and 63/100; unchanged in direction under the pre-registered sensitivity analysis of one disputed boiling point). The pattern behind the

Table 6. Nested leave-one-out Q^2 on the octanes (oct., $n = 18$) and the external nonane set (non., $n = 35$ for T_B , 34 for ΔH_{vap}). Mean: training-fold mean; Class.: best of the ten classical indices, selected inside each fold; IRLA: one-descriptor model on *IRLA*; Ridge: ridge regression on the edge-degree-pair counts; GM, LO: median over seeds 0–99 at budget 200; last column: seeds in which LO beat GM at budgets 200/500. All values are full-precision results rounded once.

Set, property	Mean	Class.	IRLA	Ridge	GM	LO	LO>GM
oct. T_B	−0.121	0.593	0.585	0.344	0.662	0.652	53/64
oct. ΔH_f	−0.121	0.746	0.593	0.654	0.831	0.689	14/24
oct. ΔH_{vap}	−0.121	0.884	0.948	0.954	0.888	0.953	100/100
oct. S	−0.121	0.920	0.741	0.744	0.919	0.905	35/29
oct. ω	−0.121	0.971	0.810	0.851	0.979	0.970	26/28
non. T_B	−0.060	0.502	0.400	0.682	0.577	0.586	58/63
non. ΔH_{vap}	−0.062	0.857	0.897	0.927	0.853	0.900	95/98

vaporization result is the same on both sets: models without the imbalance coordinate (the classical indices and GM) reach $Q^2 = 0.853$ – 0.888 , every model containing it (*IRLA*, ridge, LO) reaches $Q^2 = 0.897$ – 0.954 , and for ΔH_{vap} the fixed pure- γ point $LO(0, 0, 1)$ is the index selected in every fold on both sets when it is admitted to the fixed-index comparison. The base-lines, however, also show the limit of the claim. Ridge regression on the full degree-pair count vector predicts the nonanes better than every index model, for both properties ($Q^2 = 0.927$ for ΔH_{vap} , 0.682 for T_B); on the octanes it matches the Loyola model on ΔH_{vap} and is weaker elsewhere. This is consistent with Proposition 5 and the remark following it: the imbalance coordinate re-weights degree-pair information rather than adding to it, and its value is that one interpretable descriptor recovers most of what a model on all ten degree-pair counts achieves for the vaporization enthalpy, not that it outperforms such a model. Nor is the specific exponential form essential: the parameter-free aggregate *IRLA* comes within 0.003 of the median Loyola model on the nonanes (0.897 against 0.900), so the gain over the parent is carried by the imbalance information itself.

Finally, the nested design refits every model on the nonanes, which tests whether the *procedure* replicates, not whether an octane-trained model transfers. In an exploratory analysis added after review (not

pre-registered), the parameters selected on all 18 octanes were frozen and applied to the nonanes. With the parameters frozen and only the linear calibration refitted, the Loyola model beat the parent for ΔH_{vap} in 100/100 seeds at both budgets (median $Q^2 = 0.910$ and 0.916 against 0.864 and 0.865) and for T_B in 65/100 and 82/100. With the octane calibration frozen as well, both models extrapolate poorly from C_8 to C_9 and the Loyola model is worse than the parent (median Q^2 between -2.22 and -0.72 for LO against 0.06 to 0.44 for GM): the transferable object is the parameter direction, not a fitted octane model.

The seed counts above measure optimizer variance only. To indicate molecule-level uncertainty we also computed, post hoc, a paired bootstrap over molecules (2000 resamples) of the median-over-seeds ΔQ^2 , conditional on the fitted outer predictions (selection is not repeated within resamples, so the intervals understate total uncertainty). The 95% intervals for LO minus GM on ΔH_{vap} are $[+0.014, +0.111]$ on the octanes but $[-0.006, +0.113]$ on the nonanes at budget 200 ($[-0.005, +0.119]$ at 500): the external gain met the pre-registered criterion, yet its interval narrowly includes zero. The interval for LO minus ridge on the nonanes, $[-0.101, +0.032]$, includes zero as well, and the T_B intervals are wide ($[-0.212, +0.222]$ for LO minus GM on the nonanes). The external evidence is therefore suggestive rather than conclusive; establishing the effect would require a larger, measured dataset evaluated under a protocol fixed in advance.

5.5 Degeneracy across tree orders

Degeneracy measures an index’s ability to discriminate non-isomorphic graphs [29]; lower is better. On the 106 non-isomorphic trees of order 10 we use the distinct-value convention

$$\text{Degeneracy}(\%) = 100 \left(1 - \frac{k}{N} \right), \quad k = \#\{\text{distinct index values}\}, \quad N = 106,$$

i.e. the number $N - k$ of structures in excess of the distinct values, as a percentage of N (this is not the share of structures lacking a unique value, which is larger: for $LO(0,0,1)$ on this set 45 of the 106 trees, 42.5%,

share their value with another tree). The pure- γ points $LO(0,0,1)$ and $LO(0,0,2)$ attain degeneracy 25.5% (79 distinct values), among the lowest of all compared indices and tied with R , GA , and AG ; indeed the 106 trees realise only 79 distinct edge-degree-pair count vectors, so 25.5% is the minimum degeneracy attainable by *any* degree-pair-based index on this set, and the pure- γ points attain this floor — for vertex-degree-based weights φ_{ij} that are algebraic numbers, Rada [40] proved that the exponential index e^φ discriminates the equivalence classes of the corresponding $\hat{m}_\varphi$ -structure relation; the pure- γ imbalance weights used here are rational, hence algebraic, so that theorem separates the classes of graphs with different q -value histograms (not the degree-pair profiles themselves, since q is not injective on pairs: $q_{12} = q_{24}$ and $q_{ii} = 0$), and attainment of the 79-profile floor on this particular order-10 tree set is the separate computational observation reported here; the first Zagreb index $LO(0,1,0)$ (83.0%) and the second Zagreb index $LO(1,0,0)$ (67.0%) are the most degenerate (Figure 2).

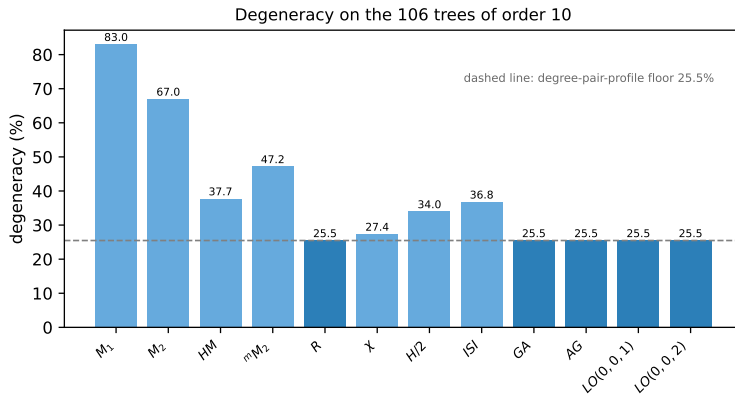


Figure 2. Degeneracy percentage of the compared indices on the 106 non-isomorphic trees of order 10. The dashed line marks the degree-pair-profile floor, $25.5\% = 100(1 - 79/106)$: the 106 trees realise only 79 distinct edge-degree-pair count vectors, so no vertex-degree-based index can fall below it.

To test whether this floor attainment is peculiar to order 10, the same computation was repeated for all non-isomorphic trees of orders 7 through

12 and, separately, for the molecular subsets with $\Delta \leq 4$ (Table 7). The pure- γ points attain the degree-pair-profile floor at *every* order in both sets. Honesty requires noting that the Randić index also attains the floor at every order in this range (while χ falls off the floor from order 10), so floor attainment distinguishes the pure- γ points from the most degenerate classical indices (M_1 , M_2) but not from R ; the observation is that the imbalance weights lose nothing, not that they are unique in this respect. Attainment on orders 7–12 is a computational observation, not a proof of injectivity on degree-pair profiles for all orders. Indeed attainment fails at larger orders: because $LO(\cdot; 0, 0, \gamma)$ depends only on the histogram of q -values, two trees with distinct degree-pair profiles but equal q -histograms receive equal values for every γ . Among the trees of order 13 the profiles $\{(1, 2)^1, (1, 3)^2, (1, 6)^5, (2, 3)^2, (2, 6)^1, (3, 3)^1\}$ and $\{(1, 3)^3, (1, 6)^5, (2, 2)^1, (2, 3)^2, (3, 6)^1\}$ (exponents are edge multiplicities) share the q -histogram $\{0^1, \frac{1}{5}^2, \frac{1}{3}^1, \frac{1}{2}^3, \frac{5}{7}^5\}$, and the pure- γ points realise 566 distinct values against 570 profiles; among molecular trees ($\Delta \leq 4$) the first failure occurs at order 16 (809 values against 810 profiles).

Table 7. Discrimination across tree orders $n = 7$ –12: number of trees N , number of distinct edge-degree-pair profiles (the floor attainable by any vertex-degree-based index), and distinct-value counts for representative indices. “mol.” denotes the molecular subset $\Delta \leq 4$.

n	set	N	floor	M_1	M_2	R	$LO(0, 0, 1)$	$LO(0, 0, 2)$
7	all	11	11	7	9	11	11	11
7	mol.	9	9	5	7	9	9	9
8	all	23	21	9	17	21	21	21
8	mol.	18	16	6	13	16	16	16
9	all	47	40	13	22	40	40	40
9	mol.	35	28	7	16	28	28	28
10	all	106	79	18	35	79	79	79
10	mol.	75	49	8	21	49	49	49
11	all	235	152	21	43	152	152	152
11	mol.	159	83	9	25	83	83	83
12	all	551	294	27	57	294	294	294
12	mol.	355	137	10	28	137	137	137

5.6 Structure sensitivity on decane isomers

The structure-sensitivity criterion for descriptor quality was introduced by Furtula, Gutman and Dehmer [15], with a fingerprint-based variant due to Rakić and Furtula [42]. We implement the published protocol on the 75 decane isomer trees Ψ : for each tree G , the similar-structure set $S(G) \subset \Psi$ consists of the non-isomorphic trees obtainable from G by relocating a single edge within the molecular class (graph edit distance two, $\Delta \leq 4$ preserved); then

$$SS(TI, G) = \frac{1}{|S(G)|} \sum_{\Omega \in S(G)} \frac{|TI(\Omega) - TI(G)|}{TI(G)},$$

$$Abr(TI, G) = \max_{\Omega \in S(G)} \frac{|TI(\Omega) - TI(G)|}{TI(G)},$$

and the class-level $SS(TI)$ and $Abr(TI)$ are the means over all 75 trees. On this class every $S(G)$ is nonempty (sizes 7 to 43), the neighbour relation is symmetric, and $Abr \geq SS$ holds throughout, as it must. As an external control, the implementation reproduces the published decane-class values of Barman and Das [6] for the seven indices common to both studies (M_1 , mM_2 , R , χ , H , GA , AG ; fourteen tabulated SS/Abr values): eleven agree exactly at the four displayed decimals and the remaining three differ by one unit in the fourth decimal (the cell-by-cell comparison is in the file `fgd_published_control.csv` of the reproducibility package; our values for all twelve compared indices are in Table 8). Under the published desideratum (SS large, Abr small [15]), the pure- γ point $LO(0, 0, 2)$ attains the largest structure sensitivity in our comparison set (the forgotten index F and the elliptic Sombor index, not included here, have larger SS on this class [6], but smaller SS/Abr ratios than either pure- γ point) ($SS = 0.1257$, marginally above the hyper-Zagreb index at 0.1256) with a markedly smaller abruptness than hyper-Zagreb (0.274 versus 0.313), and the two pure- γ points have the two largest sensitivity-to-abruptness ratios (0.4579 and 0.4565). Given the redundancy analysis of Section 5.1, the SS margin over hyper-Zagreb should be read as a tie in sensitivity at smaller abruptness, not as a ranking.

Table 8. Structure sensitivity SS , abruptness Abr , and their ratio on the 75 decane isomers under the published similar-structure protocol of [15] (single-edge relocation within the molecular class).

Index	SS	Abr	SS/Abr
M_1	0.0566	0.1336	0.4238
M_2	0.0922	0.2443	0.3776
HM	0.1256	0.3126	0.4017
mM_2	0.0516	0.1177	0.4384
R	0.0264	0.0588	0.4481
χ	0.0264	0.0591	0.4463
$H/2$	0.0516	0.1143	0.4514
ISI	0.0226	0.0597	0.3795
GA	0.0241	0.0535	0.4507
AG	0.0256	0.0579	0.4415
$LO(0, 0, 1)$	0.0624	0.1367	0.4565
$LO(0, 0, 2)$	0.1257	0.2745	0.4579

Earlier drafts of this study reported the coefficient of variation σ/μ and relative range $(\max - \min)/\mu$ over the isomer class under the names SS and Abr ; those are global dispersion summaries, not the published pairwise quantities, and they are retained only as internal artefacts of the reproducibility package under the names *normalized dispersion* and *normalized range*.

6 Conclusion and further directions

The Loyola family is the normalized degree-imbalance exponential extension of the Gutman–Milovanović family $M_{\alpha,\beta}$ [24, 35], to which it reduces identically at $\gamma = 0$. What the third parameter contributes is now stated precisely: the imbalance direction is affinely independent of the two parent directions on the admissible degree-pair set for $\Delta \geq 3$ (Proposition 2; a class-level statement, see Remark 3); the family has a globally convex logarithm whose γ -derivatives are the mean and variance of the edge imbalance under an induced edge measure (Theorem 3); and the two-sided bounds of Section 4 extend inequalities known for the parent to the $\gamma \neq 0$ regime, several of them sharp. The generic discrimination of distinct degree-pair

profiles (Proposition 5) is inherited from the parent rather than created by γ , and the imbalance coordinate re-weights the degree-pair information without adding to it.

Chemometrically the conclusions are deliberately honest. Every vertex-degree-based index on the octane benchmark operates above an irreducible floor set by two degree-pair collision pairs (Proposition 4); the twelve-index comparison set is effectively two-dimensional, so third-decimal correlation differences carry no ranking information; and at matched search budget, across one hundred random-search seeds and two budgets, the free imbalance parameter improved nested out-of-sample prediction of ΔH_{vap} in every seed of the sweep (93–96 of 100 when tetramethylbutane is excluded), usually degrades it on three other properties and is budget-dependent on the fifth (Table 5; a descriptive summary of optimizer variance on one fixed 18-molecule dataset) — it is not a systematic predictive upgrade over the optimized parent. A pre-registered external test on the 35 nonanes replicated the vaporization-enthalpy effect against compilation values (95 and 98 of 100 seeds; mostly not calorimetric determinations) and gave a mixed result for the experimentally grounded boiling point; octane-selected parameters transfer to the nonanes after recalibration, but a frozen octane model does not; the molecule-level bootstrap interval for the external vaporization gain narrowly includes zero, so this evidence is suggestive rather than conclusive; and ridge regression on the full degree-pair counts predicted the nonanes better than every index model (Section 5.4): the imbalance coordinate is a parsimonious, interpretable summary of degree-pair information, not a better predictor than a model that uses all of it. Its measurable contribution is structural: the pure- γ points attain the degree-pair discrimination floor at every tree order 7–12 tested (as does the Randić index; attainment fails from order 13, molecular trees from order 16), and under the published structure-sensitivity protocol they combine hyper-Zagreb-level sensitivity with smaller abruptness and the largest sensitivity-to-abruptness ratios in the comparison set. The Loyola family is hence best viewed as a provably independent, analytically transparent third direction on the Gutman–Milovanović parameter space, confined — like every vertex-degree-based family — to the finite-dimensional degree-

pair descriptor space on molecular graphs.

Building on the Loyola index, the following open problems indicate natural directions for further work:

1. Sharp two-sided bounds for $LO(G; \alpha, \beta, \gamma)$ via degree-based indices lying outside the family, such as the atom-bond-connectivity index ABC and the Sombor index.
2. Extremal-graph characterisations (not attempted in this paper) for several graph classes, including trees, unicyclic and bicyclic graphs.
3. Chemometric extension to additional chemical compound classes such as benzene derivatives, alkane homologues beyond the octanes, nonanes and decanes studied here and to compound-class-specific QSPR endpoints.

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A Octane isomer property data

For reproducibility, Table 9 lists the full 18-isomer dataset used in Section 5. Units: T_B in $^{\circ}\text{C}$; ΔH_f and ΔH_{vap} in kcal mol^{-1} (ΔH_{vap} at 298 K); S in $\text{cal mol}^{-1}\text{K}^{-1}$; ω dimensionless. The ΔH_{vap} values of 3,3-dimethylhexane (8.97), 3-ethyl-3-methylpentane (9.08), 2,2,3-trimethylpentane (8.83) and 2,3,3-trimethylpentane (8.90) were corrected in this version to values consistent with the NIST WebBook determinations (previously 9.04, 9.21, 8.88 and 9.02; see Section 5.1 for the exact-versus-within-uncertainty status of each). The ΔH_{vap} entry of 2,2,3,3-tetramethylbutane is a benchmark-compilation value whose physical reference state is unresolved (Section 5.1).

Table 9. Physicochemical properties of the 18 octane isomers.

Isomer	T_B	ΔH_f	ΔH_{vap}	S	ω
<i>n</i> -octane	125.6	−49.82	9.92	111.55	0.398
2-methylheptane	117.6	−51.50	9.48	109.84	0.378
3-methylheptane	118.9	−50.82	9.52	111.26	0.371
4-methylheptane	117.7	−50.69	9.48	109.32	0.372
2,2-dimethylhexane	106.8	−53.71	8.92	103.13	0.339
2,3-dimethylhexane	115.6	−51.13	9.27	108.02	0.348
2,4-dimethylhexane	109.4	−52.44	9.03	106.98	0.344
2,5-dimethylhexane	109.1	−53.21	9.05	105.72	0.357
3,3-dimethylhexane	111.9	−52.61	8.97	104.74	0.322
3,4-dimethylhexane	117.7	−50.91	9.32	106.59	0.340
3-ethyl-2-methylpentane	115.6	−50.48	9.21	106.06	0.330
3-ethyl-3-methylpentane	118.3	−51.38	9.08	101.48	0.302
3-ethylhexane	118.5	−50.40	9.48	109.43	0.362
2,2,3-trimethylpentane	109.8	−52.61	8.83	101.31	0.300
2,2,4-trimethylpentane	99.2	−53.57	8.40	101.81	0.305
2,3,3-trimethylpentane	114.8	−51.73	8.90	101.31	0.291
2,3,4-trimethylpentane	113.5	−51.97	9.01	102.39	0.317
2,2,3,3-tetramethylbutane	106.5	−53.99	8.41	93.06	0.247

References

- [1] S. Ahmad, R. Farooq, K. C. Das, The general Sombor index of extremal trees with a given maximum degree, *MATCH Commun. Math. Comput. Chem.* **94** (2025) 825–853.
- [2] M. O. Albertson, The irregularity of a graph, *Ars Combin.* **46** (1997) 219–225.
- [3] A. Ali, I. Gutman, B. Furtula, Extremal results on bond incident degree indices of graphs: a survey, *Aequationes Math.* **100** (2026) #66.
- [4] A. Ali, I. Gutman, B. Furtula, A. M. Albalahi, A. E. Hamza, On chemical and mathematical characteristics of generalized degree-based molecular descriptors, *AIMS Math.* **10** (2025) 6788–6804.
- [5] M. Arockiaraj, D. Paul, J. Clement, S. Tigga, K. Jacob, K. Balasubramanian, Novel molecular hybrid geometric-harmonic-Zagreb degree based descriptors and their efficacy in QSPR studies of polycyclic aromatic hydrocarbons, *SAR QSAR Environ. Res.* **34** (2023) 569–589.

-
- [6] J. Barman, S. Das, Geometric approach to degree-based topological index: Hyperbolic Sombor index, *MATCH Commun. Math. Comput. Chem.* **95** (2026) 63–94.
- [7] B. Bollobás, P. Erdős, Graphs of extremal weights, *Ars Combin.* **50** (1998) 225–233.
- [8] R. Cruz, J. Monsalve, J. Rada, Extremal values of vertex-degree-based topological indices of chemical trees, *Appl. Math. Comput.* **380** (2020) #125281.
- [9] K. C. Das, A. S. Çevik, I. N. Cangul, Y. Shang, On Sombor index, *Symmetry* **13** (2021) #140.
- [10] K. C. Das, S. Elumalai, S. Balachandran, Open problems on the exponential vertex-degree-based topological indices of graphs, *Discrete Appl. Math.* **293** (2021) 38–49.
- [11] K. C. Das, I. Gutman, On Sombor index of trees, *Appl. Math. Comput.* **412** (2022) #126575.
- [12] K. C. Das, S. Mondal, On exponential geometric-arithmetic index of graphs, *J. Math. Chem.* **62** (2024) 2740–2760.
- [13] P. Das, S. Mondal, Z. Raza, A. Pal, On exponential inverse symmetric division deg index of graphs, *J. Appl. Math. Comput.* **71** (Suppl. 1) (2025) 537–561.
- [14] B. Furtula, I. Gutman, A forgotten topological index, *J. Math. Chem.* **53** (2015) 1184–1190.
- [15] B. Furtula, I. Gutman, M. Dehmer, On structure-sensitivity of degree-based topological indices, *Appl. Math. Comput.* **219** (2013) 8973–8978.
- [16] W. Gao, Y. Gao, The extremal trees for exponential vertex-degree-based topological indices, *Appl. Math. Comput.* **472** (2024) #128634.
- [17] A. Granados, A. Portilla, Y. Quintana, E. Tourís, Bounds for the Gutman–Milovanović index and some applications, *J. Math. Chem.* **63** (2025) 406–418.
- [18] R. Guji, M. Wali, The Sombor index (coindex) and Lanzhou index (coindex) of some graphs, *Axioms* **14** (2025) #164.
- [19] I. Gutman, Geometric approach to degree-based topological indices: Sombor indices, *MATCH Commun. Math. Comput. Chem.* **86** (2021) 11–16.

-
- [20] I. Gutman, Some basic properties of Sombor indices, *Open J. Discr. Appl. Math.* **4** (2021) 1–3.
- [21] I. Gutman, Survey of Sombor-type degree-based topological indices, *MATCH Commun. Math. Comput. Chem.* **96** (2026) 819–837.
- [22] I. Gutman, J. Buragohain, G. G. Nyauli, Concealed relations between vertex-degree-based topological indices, *J. Math. Chem.* **64** (2026) #39.
- [23] I. Gutman, B. Furtula, M. S. Oz, Geometric approach to vertex-degree-based topological indices: Elliptic Sombor index, theory and application, *Int. J. Quantum Chem.* **124** (2024) #e27346.
- [24] I. Gutman, E. Milovanović, I. Milovanović, Beyond the Zagreb indices, *AKCE Int. J. Graphs Comb.* **17** (2020) 74–85.
- [25] I. Gutman, M. Togan, A. Yurttaş, A. S. Çevik, I. N. Cangul, Inverse problem for sigma index, *MATCH Commun. Math. Comput. Chem.* **79** (2018) 491–508.
- [26] I. Gutman, N. Trinajstić, Graph theory and molecular orbitals. Total π -electron energy of alternant hydrocarbons, *Chem. Phys. Lett.* **17** (1972) 535–538.
- [27] Z. Hu, L. Li, X. Li, D. Peng, Extremal graphs for topological index defined by a degree-based edge-weight function, *MATCH Commun. Math. Comput. Chem.* **88** (2022) 505–520.
- [28] A. R. A. Kalaam, A. B. Greeni, M. Arockiaraj, Modified reverse degree descriptors for combined topological and entropy characterizations of 2D metal organic frameworks: applications in graph energy prediction, *Front. Chem.* **12** (2024) #1470231.
- [29] E. V. Konstantinova, The discrimination ability of some topological and information distance indices for graphs of unbranched hexagonal systems, *J. Chem. Inf. Comput. Sci.* **36** (1996) 54–57.
- [30] S. Kosari, N. Dehgard, A. Khan, Lower bound on the KG-Sombor index, *Commun. Comb. Optim.* **8** (2023) 751–757.
- [31] A. Labauf, J. B. Greenshields, F. D. Rossini, Heats of formation, combustion, and vaporization of the 35 nonanes and 75 decanes, *J. Chem. Eng. Data* **6** (1961) 261–263.

-
- [32] P. J. Linstrom, W. G. Mallard (Eds.), *NIST Chemistry WebBook, NIST Standard Reference Database Number 69*, National Institute of Standards and Technology, Gaithersburg MD, doi:10.18434/T4D303 (retrieved January 2025; re-checked August–October 2026).
- [33] J. A. Méndez-Bermúdez, R. Aguilar-Sánchez, E. D. Molina, J. M. Rodríguez, Mean Sombor index, *Discr. Math. Lett.* **9** (2022) 18–25.
- [34] I. Milovanović, E. Milovanović, M. Matejić, On some mathematical properties of Sombor indices, *Bull. Int. Math. Virtual Inst.* **11** (2021) 341–353.
- [35] E. D. Molina, J. M. Rodríguez-García, J. M. Sigarreta, S. J. Torralbas Fitz, On the Gutman–Milovanović index and chemical applications, *AIMS Math.* **10** (2025) 1998–2020.
- [36] F. Movahedi, Diminished Sombor index and its relationship with topological indices, *Filomat* **39** (2025) 10519–10532.
- [37] F. Movahedi, I. Gutman, I. Redžepović, B. Furtula, Diminished Sombor index, *MATCH Commun. Math. Comput. Chem.* **95** (2026) 141–162.
- [38] G. G. Nyauli, J. Buragohain, Geometric approach to degree-based topological indices: degree-ratio Sombor indices of graphs, *MATCH Commun. Math. Comput. Chem.* **97** (2027) 135–164.
- [39] A. Portilla, J. M. Rodríguez, I. Salas Lorenzo, J. M. Sigarreta, The Gutman–Milovanović index in mathematical chemistry, *J. Math. Chem.* **63** (2025) 2068–2087.
- [40] J. Rada, Exponential vertex-degree-based topological indices and discrimination, *MATCH Commun. Math. Comput. Chem.* **82** (2019) 29–41.
- [41] J. Rada, A finite-dimensional linear framework for vertex-degree-based topological indices, *MATCH Commun. Math. Comput. Chem.* **96** (2026) 841–863.
- [42] M. Rakić, B. Furtula, A novel method for measuring the structure sensitivity of molecular descriptors, *J. Chemometrics* **33** (2019) #e3138.
- [43] M. Randić, Characterization of molecular branching, *J. Am. Chem. Soc.* **97** (1975) 6609–6615.
- [44] T. Réti, R. Sharafadini, Á. Drégelyi-Kiss, H. Haghighin, Graph irregularity indices used as molecular descriptors in QSPR studies, *MATCH Commun. Math. Comput. Chem.* **79** (2018) 509–524.

-
- [45] Y. Shang, Sombor index and degree-related properties of simplicial networks, *Appl. Math. Comput.* **419** (2022) #126881.
- [46] G. H. Shirdel, H. Rezapour, A. M. Sayadi, The hyper-Zagreb index of graph operations, *Iranian J. Math. Chem.* **4** (2013) 213–220.
- [47] J. M. Sigarreta, Extremal problems on exponential vertex-degree-based topological indices, *Math. Biosci. Eng.* **19** (2022) 6985–6995.
- [48] D. Vukičević, B. Furtula, Topological index based on the ratios of geometrical and arithmetical means of end-vertex degrees of edges, *J. Math. Chem.* **46** (2009) 1369–1376.
- [49] Z. Wang, Y. Mao, Y. Li, B. Furtula, On relations between Sombor and other degree-based indices, *J. Appl. Math. Comput.* **68** (2022) 1–17.
- [50] H. Wiener, Structural determination of paraffin boiling points, *J. Am. Chem. Soc.* **69** (1947) 17–20.
- [51] R. C. Wilhoit, B. J. Zwolinski, *Handbook of Vapor Pressures and Heats of Vaporization of Hydrocarbons and Related Compounds*, Texas A&M Research Foundation, College Station, 1971.
- [52] B. Zhou, N. Trinajstić, On a novel connectivity index, *J. Math. Chem.* **46** (2009) 1252–1270.
- [53] B. Zhou, N. Trinajstić, On general sum-connectivity index, *J. Math. Chem.* **47** (2010) 210–218.
- [54] X. Zuo, B. A. Rather, M. Imran, A. Ali, On some topological indices defined via the modified Sombor matrix, *Molecules* **27** (2022) #6772.